

XCPC Templates

Skyclad Observer

Continue

October 8, 2026

Contents

1 代数结构 2

1.1 幺半群 2

1.2 幺半群作用 3

1.3 半群快速幂 4

2 模运算 4

2.1 Barrett 模乘 4

2.2 运算 4

2.3 模数类 5

2.4 all inverse 5

3 图论 5

3.1 Base 5

3.2 最短路 6

3.2.1 Dijkstra 6

3.2.2 01-BFS 7

3.2.3 Bellman-Ford 8

3.2.4 SPFA 8

3.2.5 Floyd 8

3.2.6 路径重建 9

3.3 拓扑排序 9

3.4 欧拉通路/回路 10

3.5 最小生成树 11

3.5.1 Kruskal 11

3.5.2 Prim 11

3.5.3 Borůvka 12

3.5.4 Kruskal 重构树 12

3.6 连通性 13

3.6.1 强连通分量 13

3.6.2 边双连通分量 14

3.6.3 点双连通分量 14

3.6.4 圆方树 15

3.7 树链剖分 17

3.8 LCA 18

3.9 树上倍增 19

3.10 虚树 20

3.11 2-SAT 21

4 网络流 21

4.1 最大流最小割 21

4.2 最小费用最大流 23

5 数论 24

5.1 扩展欧几里得算法 24

5.2 线性筛 25

5.3 莫比乌斯反演 25

5.4 欧拉函数 25

5.5 Miller-Rabin 26

5.6 Pollard-Rho 27

5.7 约数构造 27

6 组合数学 28

6.1 组合数 28

6.2 康托展开 29

6.3 逆序对 29

6.4 容斥原理 30

6.5 zeta 变换 30

6.6 Polyomino 31

7 数据结构 32

7.1 线段树 32

7.1.1 线段树 32

7.1.2 懒标记线段树 33

7.1.3 动态开点线段树 35

7.2 树状数组 37

7.2.1 树状数组 37

7.2.2 区间加、单点查树状数组 37

7.2.3 区间加、区间和树状数组 38

7.2.4 二维树状数组 38

7.3 倍增 39

7.4 并查集 39

7.5 稀疏表 40

7.6 01-Trie 40

7.7 笛卡尔树 41

8 线性代数 42

8.1 矩阵 42

8.2 高斯消元 43

8.3 GF(2) 空间 44

8.3.1 异或线性基 44

8.3.2 逆矩阵 45

8.3.3 转置 46

8.3.4 GF(2) 线性方程组 46

9 字符串 47

9.1 Manacher 47

9.2 KMP 47

9.3 Trie 47

9.4 Z 函数 48

9.5 AC 自动机 48

10 杂项 49

10.1 debug 49

10.2 template 49

10.3 二分答案 50

10.4 i128 50

10.5 日期 51

1 代数结构

1.1 么半群

```

1 // Template
2 // struct Monoid {
3 //     struct Type {
4 //         i64 v;
5 //     };
6 //     static Type e() {
7 //         return {};
8 //     }
9 //     static Type op(const Type& a, const Type& b) {
10 //         return {};
11 //     }
12 //     static Type from(const i64& x) {
13 //         return {x};
14 //     }
15 // };
16
17 template <typename T>
18 struct AddMono {
19     using Type = T;
20     static Type e() {
21         return {0};
22     }
23     static Type op(const Type& a, const Type& b) {
24         return {a + b};
25     }
26     static Type from(const i64& x) {
27         return {x};
28     }
29     static Type inv(const Type& a) {
30         return {-a};
31     }
32     static Type pow(const Type& a, i64 k) {
33         return a * k;
34     }
35 };
36
37 template <typename T>
38 struct MulMono {
39     using Type = T;
40     static Type e() {

```

```

41         return {1};
42     }
43     static Type op(const Type& a, const Type& b) {
44         return {a * b};
45     }
46     static Type from(const i64& x) {
47         return {x};
48     }
49     static Type inv(const Type& a) {
50         return T{1} / a;
51     }
52 };
53
54 template <typename T>
55 struct MinMono {
56     using Type = T;
57     static Type e() {
58         return {inf<T>};
59     }
60     static Type op(const Type& a, const Type& b) {
61         return {min(a, b)};
62     }
63     static Type from(const i64& x) {
64         return {x};
65     }
66     static Type pow(const Type& a, i64 k) {
67         return (k == 0 ? e() : a);
68     }
69 };
70
71 template <typename T>
72 struct MaxMono {
73     using Type = T;
74     static Type e() {
75         return {-inf<T>};
76     }
77     static Type op(const Type& a, const Type& b) {
78         return {max(a, b)};
79     }
80     static Type from(const i64& x) {
81         return {x};
82     }
83     static Type pow(const Type& a, i64 k) {

```

```

84     return (k == 0 ? e() : a);
85 }
86 };
87
88 template <typename T>
89 struct GcdMono {
90     using Type = T;
91     static Type e() {
92         return {0};
93     }
94     static Type op(const Type& a, const Type& b) {
95         return {gcd(a, b)};
96     }
97     static Type from(const i64& x) {
98         return {x};
99     }
100    static Type pow(const Type& a, i64 k) {
101        return (k == 0 ? e() : a);
102    }
103 };
104
105 template <typename T>
106 struct LcmMono {
107     using Type = T;
108     static Type e() {
109         return {1};
110     }
111     static Type op(const Type& a, const Type& b) {
112         return {lcm(a, b)};
113     }
114     static Type from(const i64& x) {
115         return {x};
116     }
117     static Type pow(const Type& a, i64 k) {
118         return (k == 0 ? e() : a);
119     }
120 };
121
122 template <typename T>
123 struct XorMono {
124     using Type = T;
125     static Type e() {
126         return {0};

```

```

127     }
128     static Type op(const Type& a, const Type& b) {
129         return {a ^ b};
130     }
131     static Type inv(const Type& a) {
132         return {a};
133     }
134     static Type power(const Type& a, i64 k) {
135         return (k & 1) ? a : e();
136     }
137 };

```

1.2 幺半群作用

```

1 // Template
2 // struct ActMono {
3 //     using Mono = Mono;
4 //     using Tag = Tag;
5 //     using X = Mono::Type;
6 //     using Y = Tag::Type;
7 //     static X act(const X& x, const Y& y, int len) {
8 //         return {};
9 //     }
10 // };
11
12 template <typename T>
13 struct SumAdd {
14     using Mono = AddMono<T>;
15     using Tag = AddMono<T>;
16     using X = Mono::Type;
17     using Y = Tag::Type;
18     static X act(const X& x, const Y& y, int len) {
19         return {x + y * len};
20     }
21 };
22
23 template <typename T>
24 struct SumMul {
25     using Mono = AddMono<T>;
26     using Tag = MulMono<T>;
27     using X = Mono::Type;
28     using Y = Tag::Type;
29     static X act(const X& x, const Y& y, int len) {
30         return {x * y};

```

```

31     }
32 };
33
34 template <typename T>
35 struct MinAdd {
36     using Mono = MinMono<T>;
37     using Tag = AddMono<T>;
38     using X = Mono::Type;
39     using Y = Tag::Type;
40     static X act(const X& x, const Y& y, int len) {
41         return {x + y};
42     }
43 };
44
45 template <typename T>
46 struct MaxAdd {
47     using Mono = MaxMono<T>;
48     using Tag = AddMono<T>;
49     using X = Mono::Type;
50     using Y = Tag::Type;
51     static X act(const X& x, const Y& y, int len) {
52         return {x + y};
53     }
54 };

```

1.3 半群快速幂

```

1 template <typename Mono>
2 typename Mono::Type power(typename Mono::Type a, i64 b) {
3     if constexpr (requires { Mono::pow(a, b); }) {
4         return Mono::pow(a, b);
5     } else {
6         auto res = Mono::e();
7         while (b > 0) {
8             if (b & 1) {
9                 res = Mono::op(res, a);
10            }
11            a = Mono::op(a, a);
12            b >>= 1;
13        }
14        return res;
15    }
16 }

```

2 模运算

2.1 Barrett 模乘

```

1 class Barrett {
2 public:
3     constexpr Barrett(uint32_t Mod) : mod_(Mod), mu_(static_cast<uint64_t>(-1) /
Mod + 1) {}
4
5     constexpr uint32_t mod() const {
6         return mod_;
7     }
8
9     constexpr uint32_t multiply(uint32_t a, uint32_t b) const {
10         uint64_t z = static_cast<uint64_t>(a) * b;
11         uint64_t q = static_cast<uint64_t>((static_cast<__uint128_t>(z) * mu_) >>
64);
12         uint64_t y = q * mod_;
13         return static_cast<uint32_t>(z - y + (z < y ? mod_ : 0));
14     }
15 private:
16     uint32_t mod_;
17     uint64_t mu_;
18 };
19

```

2.2 运算

```

1 i64 mul(i64 a, i64 b, i64 m) {
2     return static_cast<i128>(a) * b % m;
3 }
4
5 i64 power(i64 a, i64 b, i64 m) {
6     i64 res = 1 % m;
7     while (b > 0) {
8         if (b & 1) {
9             res = mul(res, a, m);
10        }
11        a = mul(a, a, m);
12        b >>= 1;
13    }
14    return res;
15 }
16
17 template <typename T, std::integral U>
18 constexpr T power(T a, U b) {

```

```

19  assert(b >= 0);
20  T res{1};
21  while (b) {
22      if (b & 1) {
23          res *= a;
24      }
25      b >>= 1;
26      a *= a;
27  }
28  return res;
29 }

```

2.3 模数类

```

1  template <u32 P>
2  struct Mint {
3      u32 v;
4      Mint(i64 v_ = 0) : v((v_ %= P) < 0 ? v_ + P : v_) {}
5      Mint inv() const {
6          i64 x, y;
7          exgcd(v, P, x, y);
8          return x;
9      }
10     Mint& operator+=(Mint rhs) & {
11         v += rhs.v;
12         if (v >= P) v -= P;
13         return *this;
14     }
15     Mint& operator-=(Mint rhs) & {
16         v -= rhs.v;
17         if (v >= P) v += P;
18         return *this;
19     }
20     Mint& operator*=(Mint rhs) & {
21         v = u64(v) * rhs.v % P;
22         return *this;
23     }
24     Mint& operator/=(Mint rhs) & {
25         return *this *= rhs.inv();
26     }
27     friend Mint operator-(Mint rhs) { return rhs.v ? Mint(P - rhs.v) : rhs; }
28     friend Mint operator+(Mint lhs, Mint rhs) { return lhs += rhs; }
29     friend Mint operator-(Mint lhs, Mint rhs) { return lhs -= rhs; }
30     friend Mint operator*(Mint lhs, Mint rhs) { return lhs *= rhs; }

```

```

31     friend Mint operator/(Mint lhs, Mint rhs) { return lhs /= rhs; }
32     friend bool operator==(Mint lhs, Mint rhs) { return lhs.v == rhs.v; }
33     friend bool operator!=(Mint lhs, Mint rhs) { return lhs.v != rhs.v; }
34     friend istream& operator>>(istream& is, Mint& v) { i64 x; is >> x; v = x;
return is; }
35     friend ostream& operator<<(ostream& os, const Mint& v) { return os << v.v; }
36 };
37
38 constexpr u32 P = 998244353;
39 using Fp = Mint<P>;

```

2.4 all inverse

```

1  template <typename Fp>
2  vector<Fp> allInverse(const vector<Fp>& a) {
3      for (auto& v : a) {
4          assert(v != 0);
5      }
6      int n = a.size();
7      vector<Fp> res(n + 1);
8      res[0] = Fp{1};
9      for (int i = 0; i < n; i++) {
10         res[i + 1] = res[i] * a[i];
11     }
12     Fp inv = res[n].inverse();
13     res.pop_back();
14     for (int i = n - 1; i >= 0; i--) {
15         res[i] *= inv;
16         inv *= a[i];
17     }
18     return res;
19 }

```

3 图论

3.1 Base

```

1  template <typename T>
2  struct Edge {
3      int u, v;
4      T w;
5      Edge() {}
6      Edge(int u_, int v_, T w_) : u(u_), v(v_), w(w_) {}
7  };
8

```

```

9 template <typename T>
10 struct Graph {
11     vector<Edge<T>> edges;
12     vector<vector<pair<int, T>>> adj;
13     int n;
14
15     Graph() : n(0) {}
16     Graph(int n_) : n(n_) {
17         adj.resize(n);
18     }
19
20     void addEdge(int u, int v, T w = {0}) {
21         edges.emplace_back(u, v, w);
22         adj[u].emplace_back(v, w);
23     }
24 };

```

3.2 最短路

3.2.1 Dijkstra

```

1 template <typename T>
2 pair<vector<T>, vector<int>> dijkstra(vector<vector<pair<int, T>>> adj, int s) {
3     int n = adj.size();
4     vector<T> dist(n, inf<T>);
5     vector<int> pre(n, -1);
6     using P = pair<T, int>;
7     priority_queue<P, vector<P>, greater<P>> pq;
8
9     dist[s] = 0;
10    pq.emplace(0, s);
11
12    while (!pq.empty()) {
13        auto [d, u] = pq.top();
14        pq.pop();
15        if (d != dist[u]) {
16            continue;
17        }
18        for (auto& [v, w] : adj[u]) {
19            if (dist[v] > dist[u] + w) {
20                dist[v] = dist[u] + w;
21                pre[v] = u;
22                pq.emplace(dist[v], v);
23            }
24        }
25    }
26 }

```

```

25 }
26
27     return {dist, pre};
28 }
29
30 template <typename T>
31 pair<vector<T>, vector<int>> dijkstraDense(vector<vector<pair<int, T>>> adj, int s) {
32     int n = adj.size();
33     vector<T> dist(n, inf<T>);
34     vector<int> pre(n, -1);
35     vector<int> done(n, false);
36
37     dist[s] = 0;
38
39     for (int _ = 0; _ < n; _++) {
40         int u = -1;
41         for (int v = 0; v < n; v++) {
42             if (!done[v] && (u == -1 || dist[v] < dist[u])) {
43                 u = v;
44             }
45         }
46         if (u == -1 || dist[u] == inf<T>) {
47             break;
48         }
49         done[u] = true;
50         for (auto [v, w] : adj[u]) {
51             if (dist[v] > dist[u] + w) {
52                 dist[v] = dist[u] + w;
53                 pre[v] = u;
54             }
55         }
56     }
57
58     return {dist, pre};
59 }
60
61 template <typename T>
62 tuple<vector<T>, vector<int>, vector<int>> dijkstra(vector<vector<pair<int, T>>> adj, vector<int>& s) {
63     int n = adj.size();
64     vector<T> dist(n, inf<T>);
65     vector<int> pre(n, -1), root(n, -1);
66     using P = pair<T, int>;

```

```

67 priority_queue<P, vector<P>, greater<P>> pq;
68
69 for (auto& u : s) {
70     dist[u] = 0;
71     root[u] = u;
72     pq.emplace(0, u);
73 }
74
75 while (!pq.empty()) {
76     auto [d, u] = pq.top();
77     pq.pop();
78     if (d != dist[u]) {
79         continue;
80     }
81     for (auto& [v, w] : adj[u]) {
82         if (dist[v] > dist[u] + w) {
83             dist[v] = dist[u] + w;
84             pre[v] = u;
85             root[v] = root[u];
86             pq.emplace(dist[v], v);
87         }
88     }
89 }
90
91 return {dist, pre, root};
92 }

```

3.2.2 01-BFS

```

1 template <typename T>
2 pair<vector<T>, vector<int>> bfs_01(vector<vector<pair<int, T>>& adj, int s) {
3     int n = adj.size();
4     vector<T> dist(n, inf<T>);
5     vector<int> pre(n, -1);
6     deque<int> que;
7
8     dist[s] = 0;
9     que.push_front(s);
10
11     while (!que.empty()) {
12         auto u = que.front();
13         que.pop_front();
14         for (auto& [v, w] : adj[u]) {
15             if (dist[v] > dist[u] + w) {
16                 dist[v] = dist[u] + w;

```

```

17         pre[v] = u;
18         if (w == 0) {
19             que.push_front(v);
20         } else {
21             que.push_back(v);
22         }
23     }
24 }
25
26
27 return {dist, pre};
28 }
29
30 template <typename T>
31 tuple<vector<T>, vector<int>, vector<int>> bfs_01(vector<vector<pair<int, T>>& adj, vector<int>& s) {
32     int n = adj.size();
33     vector<T> dist(n, inf<T>);
34     vector<int> pre(n, -1), root(n, -1);
35     deque<int> que;
36
37     for (auto& u : s) {
38         dist[u] = 0;
39         root[u] = u;
40         que.push_front(u);
41     }
42
43     while (!que.empty()) {
44         auto u = que.front();
45         que.pop_front();
46         for (auto& [v, w] : adj[u]) {
47             if (dist[v] > dist[u] + w) {
48                 dist[v] = dist[u] + w;
49                 pre[v] = u;
50                 root[v] = root[u];
51                 if (w == 0) {
52                     que.push_front(v);
53                 } else {
54                     que.push_back(v);
55                 }
56             }
57         }
58     }

```

```

59     return {dist, pre, root};
60 }
61

```

3.2.3 Bellman-Ford

```

1 template <typename T>
2 pair<vector<T>, vector<int>> bellmanFord(vector<vector<pair<int, T>>& adj, int s)
3 {
4     int n = adj.size();
5     vector<T> dist(n, inf<T>);
6     vector<int> pre(n, -1);
7
8     dist[s] = 0;
9
10    for (int round = 1;; round++) {
11        bool upd = false;
12        for (int u = 0; u < n; u++) {
13            if (dist[u] == inf<T>) {
14                continue;
15            }
16            for (auto& [v, w] : adj[u]) {
17                if (dist[v] > dist[u] + w) {
18                    dist[v] = dist[u] + w;
19                    pre[v] = u;
20                    upd = true;
21                    if (round >= n) {
22                        return {{}, {}};
23                    }
24                }
25            }
26        }
27        if (!upd) {
28            break;
29        }
30    }
31    return {dist, pre};
32 }
33

```

3.2.4 SPFA

```

1 template <typename T>
2 pair<vector<T>, vector<int>> spfa(vector<vector<pair<int, T>>& adj, int s) {
3     int n = adj.size();

```

```

4     vector<T> dist(n, inf<T>);
5     vector<int> pre(n, -1);
6
7     dist[s] = 0;
8
9     queue<int> q;
10    vector<char> inq(n, false);
11    vector<int> cnt(n, 0);
12    q.push(s);
13    inq[s] = true;
14
15    while (!q.empty()) {
16        int u = q.front();
17        q.pop();
18        inq[u] = false;
19        for (auto& [v, w] : adj[u]) {
20            if (dist[v] > dist[u] + w) {
21                dist[v] = dist[u] + w;
22                pre[v] = u;
23                if (!inq[v]) {
24                    q.push(v);
25                    inq[v] = true;
26                    cnt[v]++;
27                    if (cnt[v] >= n) {
28                        return {{}, {}};
29                    }
30                }
31            }
32        }
33    }
34
35    return {dist, pre};
36 }

```

3.2.5 Floyd

```

1 template <typename T>
2 void floyd(vector<vector<T>>& dist) {
3     int n = dist.size();
4     for (int k = 0; k < n; k++) {
5         for (int i = 0; i < n; i++) {
6             for (int j = 0; j < n; j++) {
7                 if (dist[i][k] < inf<T> && dist[k][j] < inf<T>) {
8                     dist[i][j] = min(dist[i][j], dist[i][k] + dist[k][j]);

```

```

9         }
10    }
11 }
12 }
13 }
14
15 template <typename T>
16 vector<vector<T>> floyd(vector<vector<pair<int, T>>& adj) {
17     int n = adj.size();
18     vector dist(n, vector<T>(n, inf<T>));
19     for (int u = 0; u < n; u++) {
20         dist[u][u] = 0;
21         for (auto& [v, w] : adj[u]) {
22             dist[u][v] = min(dist[u][v], w);
23         }
24     }
25     floyd(dist);
26     return dist;
27 }
28

```

3.2.6 路径重建

```

1 vector<int> restorePath(vector<int>& pre, int t) {
2     vector<int> path = {t};
3     while (pre[path.back()] != -1) {
4         path.push_back(pre[path.back()]);
5     }
6     reverse(path.begin(), path.end());
7     return path;
8 }

```

3.3 拓扑排序

```

1 template <typename T>
2 vector<int> toposort(vector<vector<pair<int, T>>& adj) {
3     int n = adj.size();
4     vector<int> topo, deg(n, 0);
5     for (int u = 0; u < n; u++) {
6         for (auto& [v, w] : adj[u]) {
7             deg[v]++;
8         }
9     }
10    for (int u = 0; u < n; u++) {
11        if (deg[u] == 0) {

```

```

12        topo.push_back(u);
13    }
14 }
15 for (int i = 0; i < int(topo.size()); i++) {
16     int u = topo[i];
17     for (auto& [v, w] : adj[u]) {
18         if (--deg[v] == 0) {
19             topo.push_back(v);
20         }
21     }
22 }
23 return (int(topo.size()) != n ? vector<int>() : topo);
24 }
25
26 template <typename T>
27 vector<int> lexMinToposort(vector<vector<pair<int, T>>& adj) {
28     int n = adj.size();
29     vector<int> topo, deg(n, 0);
30     for (int u = 0; u < n; u++) {
31         for (auto& [v, w] : adj[u]) {
32             deg[v]++;
33         }
34     }
35     priority_queue<int, vector<int>, greater<int>> pq;
36     for (int u = 0; u < n; u++) {
37         if (deg[u] == 0) {
38             pq.push(u);
39         }
40     }
41     while (!pq.empty()) {
42         int u = pq.top();
43         pq.pop();
44         topo.push_back(u);
45         for (auto& [v, w] : adj[u]) {
46             if (--deg[v] == 0) {
47                 pq.push(v);
48             }
49         }
50     }
51     return (int(topo.size()) != n ? vector<int>() : topo);
52 }

```

3.4 欧拉通路/回路

给定起点 s ，还需要满足所有有边的顶点属于同一个连通分量，且 s 不能是孤立点。

无向图：

- 奇度点的数量一定是偶数，因此不会出现只有 1 个奇度点的情况。
- 有 0 个奇度点时，存在从任意有边顶点 s 出发的欧拉回路。
- 有 2 个奇度点时， s 必须是其中一个，存在从 s 出发的欧拉通路。
- 奇度点超过 2 个时不存在欧拉通路。

有向图：

- 若所有顶点满足 $\text{in} = \text{out}$ ，存在从任意有边顶点 s 出发的欧拉回路。
- 若恰有一个顶点满足 $\text{out} = \text{in} + 1$ ，它必须是 s ；同时恰有一个顶点满足 $\text{in} = \text{out} + 1$ ，存在从 s 出发的欧拉通路。
- 其他情况不存在欧拉通路。

```

1 // return:
2 // 0 - isn't Eulerian path
3 // 1 - is Eulerian path
4 // 2 - is Eulerian circuit
5 template <typename T>
6 int isEulerian(vector<vector<pair<int, T>>>& adj, bool directed = true, int s = -1)
7 {
8     int n = adj.size();
9     // in and out are only for directed graph
10    vector<int> in(n, 0), out(n, 0), deg(n, 0);
11    DSU dsu(n);
12    for (int u = 0; u < n; u++) {
13        for (auto& [v, w] : adj[u]) {
14            dsu.merge(u, v);
15            if (directed) {
16                out[u]++;
17                in[v]++;
18                deg[v]++;
19            }
20            deg[u]++;
21        }
22    }
23    int root = -1;
24    for (int u = 0; u < n; u++) {
25        if (deg[u] == 0) {
26            continue;
27        }
28        if (root == -1) {
29            root = u;

```

```

30    } else if (!dsu.same(root, u)) {
31        return 0;
32    }
33 }
34
35 if (s != -1 && root != -1 && !dsu.same(s, root)) {
36     return 0;
37 }
38
39 if (directed) {
40     int diff = 0;
41     for (int u = 0; u < n; u++) {
42         diff += abs(out[u] - in[u]);
43     }
44     if (diff == 0) {
45         return 2;
46     } else if (diff == 2) {
47         if (s == -1) {
48             return 1;
49         } else {
50             return (out[s] == in[s] + 1);
51         }
52     } else {
53         return 0;
54     }
55 } else {
56     int odd = 0;
57     for (int u = 0; u < n; u++) {
58         odd += (deg[u] & 1);
59     }
60     if (odd == 0) {
61         return 2;
62     } else if (odd == 2) {
63         if (s == -1) {
64             return 1;
65         } else {
66             return (deg[s] & 1);
67         }
68     } else {
69         return 0;
70     }
71 }
72 }

```

```

73
74 template <typename T>
75 bool isEulerianPath(vector<vector<pair<int, T>>& adj, bool directed = true, int s
= -1) {
76     return isEulerian(adj, directed, s) >= 1;
77 }
78 template <typename T>
79 bool isEulerianCircuit(vector<vector<pair<int, T>>& adj, bool directed = true, int
s = -1) {
80     return isEulerian(adj, directed, s) == 2;
81 }
82
83 template <typename T>
84 pair<vector<int>, vector<Edge<T>>> findEulerian(vector<vector<pair<int, T>>& adj,
int s = -1) {
85     // TODO
86 }

```

3.5 最小生成树

3.5.1 Kruskal

```

1 template <typename T>
2 pair<vector<Edge<T>>, T> kruskal(vector<Edge<T>>& edges, int n) {
3     T cost = 0;
4     vector<int> ord(edges.size());
5     iota(ord.begin(), ord.end(), 0);
6     sort(ord.begin(), ord.end(), [&](int i, int j) {
7         return edges[i].w < edges[j].w;
8     });
9
10    DSU dsu(n);
11    vector<Edge<T>> res;
12
13    for (auto& id : ord) {
14        auto& e = edges[id];
15        if (dsu.merge(e.u, e.v)) {
16            res.push_back(e);
17            cost += e.w;
18        }
19    }
20
21    return {res, cost};
22 }
23

```

3.5.2 Prim

```

1 template <typename T, typename F>
2 vector<Edge<T>> prim(int n, F cost) {
3     vector<Edge<T>> res;
4     vector<T> weight(n, inf<T>);
5     vector<int> pre(n);
6
7     auto add = [&](int u) -> void {
8         for (int v = 0; v < n; v++) {
9             if (pre[v] == -1) {
10                 continue;
11             }
12             T w = cost(u, v);
13             if (weight[v] > w) {
14                 weight[v] = w;
15                 pre[v] = u;
16             }
17         }
18         weight[u] = inf<T>;
19         pre[u] = -1;
20     };
21
22    add(0);
23    for (int i = 0; i < n - 1; i++) {
24        int u = min_element(weight.begin(), weight.end()) - weight.begin();
25        res.emplace_back(pre[u], u, weight[u]);
26        add(u);
27    }
28
29    return res;
30 }
31
32 template <typename T>
33 vector<Edge<T>> prim(vector<vector<pair<int, T>>& adj) {
34     int n = adj.size();
35     vector<Edge<T>> res;
36     vector<T> weight(n, inf<T>);
37     vector<int> pre(n);
38
39    auto add = [&](int u) -> void {
40        for (auto& [v, w] : adj[u]) {
41            if (pre[v] == -1) {
42                continue;
43            }

```

```

44     if (weight[v] > w) {
45         weight[v] = w;
46         pre[v] = u;
47     }
48 }
49 weight[u] = inf<T>;
50 pre[u] = -1;
51 };
52
53 add(0);
54 for (int i = 0; i < n - 1; i++) {
55     int u = min_element(weight.begin(), weight.end()) - weight.begin();
56     res.emplace_back(pre[u], u, weight[u]);
57     add(u);
58 }
59
60 return res;
61 }

```

3.5.3 Borůvka

```

1 template <typename T>
2 pair<vector<Edge<T>>, T> boruvka(vector<Edge<T>>& edges, int n) {
3     T cost = 0;
4     vector<Edge<T>> res;
5     DSU dsu(n);
6     int cnt = n;
7
8     while (cnt > 1) {
9         vector<int> best(n, -1);
10        int m = edges.size();
11
12        for (int i = 0; i < m; i++) {
13            auto& e = edges[i];
14            int u = dsu.find(e.u);
15            int v = dsu.find(e.v);
16            if (u == v) {
17                continue;
18            }
19            if (best[u] == -1 || e.w < edges[best[u]].w) {
20                best[u] = i;
21            }
22            if (best[v] == -1 || e.w < edges[best[v]].w) {
23                best[v] = i;

```

```

24        }
25    }
26
27    bool upd = false;
28    for (int u = 0; u < n; u++) {
29        if (best[u] == -1) {
30            continue;
31        }
32        auto& e = edges[best[u]];
33        if (dsu.merge(e.u, e.v)) {
34            res.push_back(e);
35            cost += e.w;
36            cnt--;
37            upd = true;
38        }
39    }
40    if (!upd) {
41        break;
42    }
43 }
44
45 return {res, cost};
46 }

```

3.5.4 Kruskal 重构树

```

1 template <typename T>
2 struct KruskalTree {
3     int n, root;
4     vector<i64> val;
5     vector<Edge<T>> edges;
6     vector<vector<int>> adj;
7
8     KruskalTree(int n_) {
9         init(n_);
10    }
11
12    void init(int n_) {
13        n = n_;
14        root = n - 1;
15        val.assign(2 * n - 1, 0);
16        adj.assign(2 * n - 1, vector<int>{});
17        edges.clear();
18    }

```

```

19 void addEdge(int u, int v, i64 w) {
20     edges.emplace_back(u, v, w);
21 }
22
23 void build() {
24     DSU dsu(2 * n - 1);
25     vector<int> ord(2 * n - 1);
26     iota(ord.begin(), ord.end(), 0);
27     sort(ord.begin(), ord.end(), [&](int x, int y) {
28         return edges[x].w < edges[y].w;
29     });
30
31     for (auto& id : ord) {
32         auto& e = edges[id];
33         int u = e.u, v = e.v;
34         i64 w = e.w;
35
36         if (!dsu.same(u, v)) {
37             u = dsu.find(u);
38             v = dsu.find(v);
39             val[++root] = w;
40
41             adj[root].push_back(u);
42             adj[root].push_back(v);
43             adj[u].push_back(root);
44             adj[v].push_back(root);
45
46             dsu.f[u] = root;
47             dsu.f[v] = root;
48             dsu.siz[root] = dsu.siz[u] + dsu.siz[v];
49
50             if (root == 2 * n - 2) {
51                 return;
52             }
53         }
54     }
55 }
56
57 };

```

3.6 连通性

3.6.1 强连通分量

```

1 struct SCC {
2     int n;

```

```

3     vector<vector<int>>> adj;
4     vector<int> stk;
5     vector<int> dfn, low, bel;
6     int cur, cnt;
7
8     SCC() {}
9     SCC(int n_) {
10         init(n_);
11     }
12
13     void init(int n_) {
14         n = n_;
15         adj.assign(n, {});
16         dfn.assign(n, -1);
17         low.resize(n);
18         bel.assign(n, -1);
19         stk.clear();
20         cur = cnt = 0;
21     }
22
23     void addEdge(int u, int v) {
24         adj[u].push_back(v);
25     }
26
27     void dfs(int u) {
28         dfn[u] = low[u] = cur++;
29         stk.push_back(u);
30
31         for (auto& v : adj[u]) {
32             if (dfn[v] == -1) {
33                 dfs(v);
34                 low[u] = min(low[u], low[v]);
35             } else if (bel[v] == -1) {
36                 low[u] = min(low[u], dfn[v]);
37             }
38         }
39
40         if (dfn[u] == low[u]) {
41             int y;
42             do {
43                 y = stk.back();
44                 bel[y] = cnt;
45                 stk.pop_back();

```

```

46         } while (y != u);
47         cnt++;
48     }
49 }
50
51 vector<int> work() {
52     for (int i = 0; i < n; i++) {
53         if (dfn[i] == -1) {
54             dfs(i);
55         }
56     }
57     for (auto& x : bel) {    // 把逆拓扑序转成拓扑序
58         x = cnt - 1 - x;
59     }
60     return bel;
61 }
62 };

```

3.6.2 边双连通分量

```

1 struct EBCC {
2     int n;
3     vector<vector<pair<int, int>>> adj;
4     vector<int> stk;
5     vector<int> dfn, low, bel;
6     int cur, cnt, ecnt;
7
8     EBCC() {}
9     EBCC(int n_) {
10         init(n_);
11     }
12
13     void init(int n_) {
14         n = n_;
15         adj.assign(n, {});
16         dfn.assign(n, -1);
17         low.resize(n);
18         bel.assign(n, -1);
19         stk.clear();
20         cur = cnt = ecnt = 0;
21     }
22
23     void addEdge(int u, int v) {
24         adj[u].emplace_back(v, ecnt);

```

```

25         adj[v].emplace_back(u, ecnt);
26         ecnt++;
27     }
28
29     void dfs(int u, int p) {
30         dfn[u] = low[u] = cur++;
31         stk.push_back(u);
32
33         for (auto& [v, eid] : adj[u]) {
34             if (eid == p) {
35                 continue;
36             }
37             if (dfn[v] == -1) {
38                 dfs(v, eid);
39                 low[u] = min(low[u], low[v]);
40             } else {
41                 low[u] = min(low[u], dfn[v]);
42             }
43         }
44
45         if (dfn[u] == low[u]) {
46             int y;
47             do {
48                 y = stk.back();
49                 bel[y] = cnt;
50                 stk.pop_back();
51             } while (y != u);
52             cnt++;
53         }
54     }
55
56     vector<int> work() {
57         for (int i = 0; i < n; i++) {
58             if (dfn[i] == -1) {
59                 dfs(i, -1);
60             }
61         }
62         return bel;
63     }
64 };

```

3.6.3 点双连通分量

```

1 struct VBCC {

```

```

2  int n;
3  vector<pair<int, int>> edges;
4  vector<vector<pair<int, int>>> adj;
5  vector<int> stk;
6  vector<int> dfn, low, cut;
7  vector<vector<int>> comp;
8  int cur;
9
10 VBCC() {}
11 VBCC(int n_) {
12     init(n_);
13 }
14
15 void init(int n_) {
16     n = n_;
17     adj.assign(n, {});
18     edges.clear();
19     dfn.assign(n, -1);
20     low.resize(n);
21     stk.clear();
22     comp.clear();
23     cut.assign(n, false);
24     cur = 0;
25 }
26
27 void addEdge(int u, int v) {
28     adj[u].emplace_back(v, edges.size());
29     adj[v].emplace_back(u, edges.size());
30     edges.emplace_back(u, v);
31 }
32
33 void dfs(int u, int p) {
34     dfn[u] = low[u] = cur++;
35     int child = 0;
36     for (auto& [v, eid] : adj[u]) {
37         if (eid == p) {
38             continue;
39         }
40         if (dfn[v] == -1) {
41             child++;
42             stk.push_back(eid);
43             dfs(v, eid);
44             low[u] = min(low[u], low[v]);

```

```

45         if (low[v] >= dfn[u]) {
46             if (p != -1) {
47                 cut[u] = true;
48             }
49             vector<int> b;
50             while (true) {
51                 int e = stk.back();
52                 stk.pop_back();
53                 b.push_back(e);
54                 if (e == eid) {
55                     break;
56                 }
57             }
58             comp.push_back(b);
59         }
60     } else if (dfn[v] < dfn[u]) {
61         stk.push_back(eid);
62         low[u] = min(low[u], dfn[v]);
63     }
64 }
65 if (p == -1 && child >= 2) {
66     cut[u] = true;
67 }
68 }
69
70 void work() {
71     for (int i = 0; i < n; i++) {
72         if (dfn[i] == -1) {
73             dfs(i, -1);
74         }
75     }
76 }
77 };

```

3.6.4 圆方树

```

1 struct BlockCutTree {
2     int n;
3     vector<pair<int, int>> edges;
4     vector<vector<pair<int, int>>> adj;
5     vector<int> stk;
6     vector<int> dfn, low, cut;
7     vector<vector<int>> comp;
8     vector<vector<int>> tree;

```

```

9  int cur;
10 int tot;

11
12 BlockCutTree() {}
13 BlockCutTree(int n_) {
14     init(n_);
15 }
16
17 void init(int n_) {
18     n = n_;
19     adj.assign(n, {});
20     edges.clear();
21     dfn.assign(n, -1);
22     low.resize(n);
23     stk.clear();
24     comp.clear();
25     cut.assign(n, false);
26     tree.assign(n, {});
27     tot = n;
28     cur = 0;
29 }
30
31 void addEdge(int u, int v) {
32     adj[u].emplace_back(v, edges.size());
33     adj[v].emplace_back(u, edges.size());
34     edges.emplace_back(u, v);
35 }
36
37 void addBlock(vector<int>& b) {
38     int x = tot++;
39     tree.emplace_back();
40     vector<int> pts;
41     for (auto& e : b) {
42         auto [u, v] = edges[e];
43         pts.push_back(u);
44         pts.push_back(v);
45     }
46     sort(pts.begin(), pts.end());
47     pts.erase(unique(pts.begin(), pts.end(), pts.end()), pts.end());
48     for (int u : pts) {
49         tree[u].push_back(x);
50         tree[x].push_back(u);
51     }

```

```

52 }
53
54 void dfs(int u, int p) {
55     dfn[u] = low[u] = cur++;
56     int child = 0;
57     for (auto& [v, eid] : adj[u]) {
58         if (eid == p) {
59             continue;
60         }
61         if (dfn[v] == -1) {
62             child++;
63             stk.push_back(eid);
64             dfs(v, eid);
65             low[u] = min(low[u], low[v]);
66             if (low[v] >= dfn[u]) {
67                 if (p != -1) {
68                     cut[u] = true;
69                 }
70                 vector<int> b;
71                 while (true) {
72                     int e = stk.back();
73                     stk.pop_back();
74                     b.push_back(e);
75                     if (e == eid) {
76                         break;
77                     }
78                 }
79                 comp.push_back(b);
80                 addBlock(b);
81             }
82             } else if (dfn[v] < dfn[u]) {
83                 stk.push_back(eid);
84                 low[u] = min(low[u], dfn[v]);
85             }
86         }
87         if (p == -1 && child >= 2) {
88             cut[u] = true;
89         }
90     }
91
92 void build() {
93     for (int i = 0; i < n; i++) {
94         if (dfn[i] == -1) {

```

```

95     dfs(i, -1);
96 }
97 }
98 }
99 };

```

3.7 树链剖分

```

1 struct HLD {
2     int n;
3     vector<int> siz, top, dep, fa, in, out, seq;
4     vector<vector<int>> adj;
5     int cur;
6
7     HLD() {}
8     HLD(int n_) {
9         n = n_;
10        siz.resize(n);
11        top.resize(n);
12        dep.resize(n);
13        fa.resize(n);
14        in.resize(n);
15        out.resize(n);
16        seq.resize(n);
17        cur = 0;
18        adj.assign(n, {});
19    }
20
21    void addEdge(int u, int v) {
22        adj[u].push_back(v);
23        adj[v].push_back(u);
24    }
25
26    void work(int root = 0) {
27        top[root] = root;
28        dep[root] = 0;
29        fa[root] = -1;
30        dfs1(root);
31        dfs2(root);
32    }
33
34    void dfs1(int u) {
35        if (fa[u] != -1) {
36            adj[u].erase(find(adj[u].begin(), adj[u].end(), fa[u]));

```

```

37        }
38
39        siz[u] = 1;
40        for (auto& v : adj[u]) {
41            fa[v] = u;
42            dep[v] = dep[u] + 1;
43            dfs1(v);
44            siz[u] += siz[v];
45            if (siz[v] > siz[adj[u][0]]) {
46                swap(v, adj[u][0]);
47            }
48        }
49    }
50
51    void dfs2(int u) {
52        in[u] = cur++;
53        seq[in[u]] = u;
54        for (auto& v : adj[u]) {
55            top[v] = (v == adj[u][0] ? top[u] : v);
56            dfs2(v);
57        }
58        out[u] = cur;
59    }
60
61    int lca(int u, int v) {
62        while (top[u] != top[v]) {
63            if (dep[top[u]] > dep[top[v]]) {
64                u = fa[top[u]];
65            } else {
66                v = fa[top[v]];
67            }
68        }
69        return dep[u] < dep[v] ? u : v;
70    }
71
72    int dist(int u, int v) {
73        return dep[u] + dep[v] - 2 * dep[lca(u, v)];
74    }
75
76    int jump(int u, int k) {
77        if (dep[u] < k) {
78            return -1;
79        }

```

```

80     int d = dep[u] - k;
81     while (dep[top[u]] > d) {
82         u = fa[top[u]];
83     }
84     return seq[in[u] - dep[u] + d];
85 }
86
87 bool isAncestor(int u, int v) {
88     return in[u] <= in[v] && in[v] < out[u];
89 }
90
91 int rootedParent(int u, int v) {
92     swap(u, v);
93     if (u == v) {
94         return u;
95     }
96     if (!isAncestor(u, v)) {
97         return fa[u];
98     }
99     auto it = upper_bound(adj[u].begin(), adj[u].end(), v, [&](int x, int y)
100 { return in[x] < in[y]; }) - 1;
101     return *it;
102 }
103
104 int rootedSize(int u, int v) {
105     if (u == v) {
106         return n;
107     }
108     if (!isAncestor(v, u)) {
109         return siz[v];
110     }
111     return n - siz[rootedParent(u, v)];
112 }
113
114 int rootedLca(int a, int b, int c) {
115     return lca(a, b) ^ lca(b, c) ^ lca(c, a);
116 }
117
118 };

```

3.8 LCA

```

1 struct LCA {
2     int logn, cur;
3     vector<int> dep, dfn;

```

```

4     vector<vector<int>>> fa;
5
6     LCA(vector<vector<int>>& adj, int root = 0) {
7         init(adj, root);
8     }
9
10    void init(vector<vector<int>>& adj, int root) {
11        int n = adj.size();
12        cur = 0;
13        logn = __lg(n);
14        fa.assign(n, vector<int>(logn + 1, -1));
15        dep.assign(n, 0);
16        dfn.assign(n, -1);
17
18        auto dfs = [&](auto&& self, int u, int p) -> void {
19            fa[u][0] = p;
20            dfn[u] = cur++;
21            for (int i = 1; i <= logn; i++) {
22                fa[u][i] = fa[fa[u][i - 1]][i - 1];
23            }
24            for (auto& v : adj[u]) {
25                if (v == p) {
26                    continue;
27                }
28                dep[v] = dep[u] + 1;
29                self(self, v, u);
30            }
31        };
32
33        dfs(dfs, root, root);
34
35        for (int i = n - 1; i >= 0; i--) {
36            if (fa[i][0] == -1) {
37                dfs(dfs, i, i);
38            }
39        }
40    }
41
42    int get(int u, int v) {
43        if (dep[u] < dep[v]) {
44            swap(u, v);
45        }
46        for (int i = logn; i >= 0; i--) {

```

```

47         if (dep[fa[u][i]] >= dep[v]) {
48             u = fa[u][i];
49         }
50     }
51     if (u == v) {
52         return u;
53     }
54     for (int i = logn; i >= 0; i--) {
55         if (fa[u][i] != fa[v][i]) {
56             u = fa[u][i];
57             v = fa[v][i];
58         }
59     }
60     return fa[u][0];
61 }
62
63 int dist(int u, int v) {
64     int res = dep[u] + dep[v] - 2 * dep[get(u, v)];
65     return res;
66 }
67 };
68
69 struct FastLCA {
70     struct DfnMin {
71         struct Type {
72             int dfn, fa;
73         };
74         static Type op(const Type& a, const Type& b) {
75             return a.dfn < b.dfn ? a : b;
76         }
77     };
78
79     int cur;
80     vector<int> dep, dfn, fa;
81     SparseTable<DfnMin> st;
82
83     FastLCA() {}
84     FastLCA(vector<vector<int>>& adj, int root = 0) {
85         init(adj, root);
86     }
87
88     void init(vector<vector<int>>& adj, int root) {
89         int n = adj.size();

```

```

90         cur = 0;
91         dep.assign(n, 0);
92         dfn.assign(n, -1);
93         fa.assign(n, -1);
94         vector<DfnMin::Type> seq(n);
95         auto dfs = [&](auto&& self, int u, int p) -> void {
96             fa[u] = p;
97             dfn[u] = cur++;
98             seq[dfn[u]] = {dfn[p], p};
99             for (auto& v : adj[u]) {
100                 if (v == p) {
101                     continue;
102                 }
103                 dep[v] = dep[u] + 1;
104                 self(self, v, u);
105             }
106         };
107         dfs(dfs, root, root);
108         for (int i = n - 1; i >= 0; i--) {
109             if (fa[i] == -1) {
110                 dfs(dfs, i, i);
111             }
112         }
113         st.init(seq);
114     }
115
116     int query(int u, int v) {
117         if (u == v) {
118             return u;
119         }
120         if (dfn[u] > dfn[v]) {
121             swap(u, v);
122         }
123         return st.query(dfn[u] + 1, dfn[v] + 1).fa;
124     }
125
126     int dist(int u, int v) {
127         return dep[u] + dep[v] - 2 * dep[query(u, v)];
128     }
129 };

```

3.9 树上倍增

```
1 template <typename Mono>
```

```

2 struct TreeLifting {
3     using X = typename Mono::Type;
4
5     vector<int> dep;
6     BinaryLifting<Mono> lift;
7
8     TreeLifting() {}
9
10    template<typename T>
11    TreeLifting(vector<vector<pair<int, T>>& adj, int root = 0) {
12        init(adj, root);
13    }
14
15    template <typename T>
16    void init(vector<vector<pair<int, T>>& adj, int root) {
17        int n = adj.size();
18        dep.assign(n, 0);
19        lift.init(n);
20        auto dfs = [&](auto&& self, int u, int p) -> void {
21            for (auto& [v, w] : adj[u]) {
22                if (v == p) {
23                    continue;
24                }
25                dep[v] = dep[u] + 1;
26                lift.add(v, u, Mono::from(w));
27                self(self, v, u);
28            }
29        };
30        lift.add(root, -1, Mono::e());
31        dfs(dfs, root, -1);
32        lift.build();
33    }
34
35    int lca(int u, int v) {
36        if (dep[u] < dep[v]) {
37            swap(u, v);
38        }
39        u = lift.jump(u, dep[u] - dep[v]);
40        if (u == v) {
41            return u;
42        }
43        for (int k = lift.logn; k >= 0; k--) {
44            if (lift.nxt[k][u] != lift.nxt[k][v]) {

```

```

45                u = lift.nxt[k][u];
46                v = lift.nxt[k][v];
47            }
48        }
49        return lift.nxt[0][u];
50    }
51
52    pair<int, X> query(int u, int v) {
53        int l = lca(u, v);
54        X res = Mono::op(lift.query(u, dep[u] - dep[l]).second, lift.query(v,
55        dep[v] - dep[l]).second);
56        return {l, res};
57    };

```

3.10 虚树

```

1 template <typename T>
2 struct VirtualTree {
3     LCA& lca;
4     vector<int> nodes;
5     vector<vector<pair<int, T>>& adj;
6
7     VirtualTree(LCA& lca) : lca(lca) {}
8
9     pair<vector<int>&, vector<vector<pair<int, T>>&> build(const vector<int>& h) {
10         nodes = h;
11         auto cmp = [&](int u, int v) { return lca.dfn[u] < lca.dfn[v]; };
12         sort(nodes.begin(), nodes.end(), cmp);
13         int m = nodes.size();
14         for (int i = 1; i < m; i++) {
15             nodes.push_back(lca.get(nodes[i - 1], nodes[i]));
16         }
17         sort(nodes.begin(), nodes.end(), cmp);
18         nodes.erase(unique(nodes.begin(), nodes.end(), nodes.end()));
19         int n = nodes.size();
20         adj.resize(n);
21         for (auto& e : adj) {
22             e.clear();
23         }
24         for (int i = 0; i < n - 1; i++) {
25             int p = lca.get(nodes[i], nodes[i + 1]);
26             int u = lower_bound(nodes.begin(), nodes.end(), p, cmp) -
27             nodes.begin();
28             int v = i + 1;

```

```

28     T w = lca.dep[nodes[v]] - lca.dep[p];
29     adj[u].emplace_back(v, w);
30     adj[v].emplace_back(u, w);
31 }
32
33 return {nodes, adj};
34 }
35 };

```

3.11 2-SAT

addClause(u, f, v, g) 加入一个析取子句:

$$(x_u = f) \vee (x_v = g).$$

这里 x_i 是第 i 个布尔变量, 变量编号从 0 开始。f 和 g 决定文字的正负: true 表示正文字 x_i , false 表示负文字 $\neg x_i$ 。

例如: addClause(3, true, 5, false) 表示

$$x_3 \vee \neg x_5.$$

一个子句 $(A \vee B)$ 等价于两条蕴含边:

$$(\neg A) \rightarrow B, (\neg B) \rightarrow A.$$

所以 addClause(u, f, v, g) 实际加入:

$$(x_u = \neg f) \rightarrow (x_v = g), (x_v = \neg g) \rightarrow (x_u = f).$$

图中的编号约定是: $2i$ 表示变量 x_i 取假值, $2i+1$ 表示变量 x_i 取真值。因此第一条边的起点是第一个文字的否定, 终点是第二个文字本身; 另一条边同理。

```

1 struct TwoSat {
2     int n;
3     SCC scc;
4     vector<int> ans;
5
6     TwoSat(int n_) : n(n_), scc(2 * n), ans(n) {}
7
8     // add clause: (x_u = f) ∨ (x_v = g)
9     void addClause(int u, bool f, int v, bool g) {
10         scc.addEdge(2 * u + !f, 2 * v + g);
11         scc.addEdge(2 * v + !g, 2 * u + f);
12     }
13
14     // (x_u = f)
15     void set(int u, bool f) {
16         addClause(u, f, u, f);
17     }
18
19     // (x_u = f) -> (x_v = g)
20     void implies(int u, int f, int v, int g) {

```

```

21         addClause(u, !f, v, g);
22     }
23
24     pair<bool, vector<int>> satisfiable() {
25         auto id = scc.work();
26         for (int i = 0; i < n; i++) {
27             if (id[2 * i] == id[2 * i + 1]) {
28                 return {false, {}};
29             }
30             ans[i] = id[2 * i] < id[2 * i + 1];
31         }
32         return {true, ans};
33     }
34 };

```

4 网络流

4.1 最大流最小割

```

1 template <typename T>
2 struct MaxFlow {
3     struct Edge_ {
4         int to;
5         T cap;
6         Edge_(int to_, T cap_) : to(to_), cap(cap_) {}
7     };
8
9     int n;
10    vector<Edge_> e;
11    vector<vector<int>> adj;
12    vector<int> cur, level;
13
14    MaxFlow() {}
15    MaxFlow(int n_) {
16        init(n_);
17    }
18
19    void init(int n_) {
20        n = n_;
21        e.clear();
22        adj.assign(n_, {});
23        cur.resize(n_);
24        level.resize(n_);
25    }
26

```

```

27 bool bfs(int s, int t) {
28     level.assign(n, -1);
29     queue<int> q;
30     level[s] = 0;
31     q.push(s);
32     while (!q.empty()) {
33         const int u = q.front();
34         q.pop();
35         for (auto& i : adj[u]) {
36             auto [v, c] = e[i];
37             if (c > 0 && level[v] == -1) {
38                 level[v] = level[u] + 1;
39                 if (v == t) {
40                     return true;
41                 }
42                 q.push(v);
43             }
44         }
45     }
46     return false;
47 }
48
49 T dfs(int u, int t, T f) {
50     if (u == t) {
51         return f;
52     }
53     auto r = f;
54     for (int& i = cur[u]; i < int(adj[u].size()); i++) {
55         const int& j = adj[u][i];
56         auto [v, c] = e[j];
57         if (c > 0 && level[v] == level[u] + 1) {
58             auto a = dfs(v, t, min(r, c));
59             e[j].cap -= a;
60             e[j ^ 1].cap += a;
61             r -= a;
62             if (r == 0) {
63                 return f;
64             }
65         }
66     }
67     return f - r;
68 }
69

```

```

70 void addEdge(int u, int v, T c) {
71     adj[u].push_back(e.size());
72     e.emplace_back(v, c);
73     adj[v].push_back(e.size());
74     e.emplace_back(u, 0);
75 }
76
77 T flow(int s, int t) {
78     T ans = 0;
79     while (bfs(s, t)) {
80         cur.assign(n, 0);
81         ans += dfs(s, t, numeric_limits<T>::max());
82     }
83     return ans;
84 }
85
86 vector<bool> minCut() {
87     vector<bool> c(n);
88     for (int i = 0; i < n; i++) {
89         c[i] = (level[i] != -1);
90     }
91     return c;
92 }
93
94 struct Edge {
95     int from;
96     int to;
97     T cap;
98     T flow;
99 };
100
101 vector<Edge> edges() {
102     vector<Edge> a;
103     for (int i = 0; i < e.size(); i += 2) {
104         Edge x;
105         x.from = e[i + 1].to;
106         x.to = e[i].to;
107         x.cap = e[i].cap + e[i + 1].cap;
108         x.flow = e[i + 1].cap;
109         a.push_back(x);
110     }
111     return a;
112 }

```

```
113 };
```

4.2 最小费用最大流

最小费用流的每次增广都要在残量网络上找最短路，而残量网络里有反向边、费用是负的，Dijkstra 直接用不了。势能数组 h 就是用来把边权「平移」成非负的。

做法是把每条边的费用换成约化费用

$$c'(u, v) = c(u, v) + h(u) - h(v),$$

并且每轮求完最短路后令 $h(v) \leftarrow h(v) + \text{dist}'(v)$ ，其中 dist' 是用约化费用算出的最短路。这样之后的约化费用依然非负——由三角不等式 $\text{dist}'(v) \leq \text{dist}'(u) + c'(u, v)$ 整理一下，正好就是新的 $c'(u, v) \geq 0$ 。

代价是路径长度被平移了常量 $h(s) - h(t)$ ，但它不影响谁是最短路。代码里每轮累加的是 $h(t)$ ：因为 $h(s)$ 恒为 0，更新后的 $h(t)$ 恰好等于原网络中 $s \rightarrow t$ 的真实最短费用。

初始势能默认全 0，这就要求**初始边权全部非负**；如果原图本来就有负费用边，得根据图的特性自行构建 h 或者直接先跑一遍 SPFA 得到初始 h ，再从参数传进来。

```
1 template<typename T>
2 struct MinCostFlow {
3     struct Edge_ {
4         int to;
5         T cap;
6         T cost;
7         Edge_(int to_, T cap_, T cost_) : to(to_), cap(cap_), cost(cost_) {}
8     };
9
10    int n;
11    vector<Edge_> e;
12    vector<vector<int>> adj;
13    vector<T> h, dist;
14    vector<int> pre;
15
16    bool dijkstra(int s, int t) {
17        dist.assign(n, numeric_limits<T>::max());
18        pre.assign(n, -1);
19        priority_queue<pair<T, int>, vector<pair<T, int>>, greater<pair<T, int>>>
20    que;
21        dist[s] = 0;
22        que.emplace(0, s);
23        while (!que.empty()) {
24            auto [d, u] = que.top();
25            que.pop();
26            if (dist[u] != d) {
27                continue;
28            }
29            for (auto& i : adj[u]) {
```

```
29            int v = e[i].to;
30            if (e[i].cap > 0 && dist[v] > d + h[u] - h[v] + e[i].cost) {
31                dist[v] = d + h[u] - h[v] + e[i].cost;
32                pre[v] = i;
33                que.emplace(dist[v], v);
34            }
35        }
36    }
37    return dist[t] != numeric_limits<T>::max();
38 }
39
40 MinCostFlow() {}
41 MinCostFlow(int n_) {
42     init(n_);
43 }
44
45 void init(int n_) {
46     n = n_;
47     e.clear();
48     adj.assign(n, {});
49 }
50
51 void addEdge(int u, int v, T cap, T cost) {
52     adj[u].push_back(e.size());
53     e.emplace_back(v, cap, cost);
54     adj[v].push_back(e.size());
55     e.emplace_back(u, 0, -cost);
56 }
57
58 void set(const vector<T>& potential) {
59     h = potential;
60 }
61
62 pair<T, T> flow(int s, int t, T need = numeric_limits<T>::max()) {
63     T flow = 0;
64     T cost = 0;
65     while (flow < need && dijkstra(s, t)) {
66         for (int i = 0; i < n; ++i) {
67             h[i] += dist[i];
68         }
69         T aug = numeric_limits<T>::max();
70         for (int i = t; i != s; i = e[pre[i]].to) {
71             aug = min(aug, e[pre[i]].cap);
```

```

72     }
73     for (int i = t; i != s; i = e[pre[i] ^ 1].to) {
74         e[pre[i]].cap -= aug;
75         e[pre[i] ^ 1].cap += aug;
76     }
77     flow += aug;
78     cost += aug * h[t];
79 }
80 return {flow, cost};
81 }
82
83 struct Edge {
84     int from;
85     int to;
86     T cap;
87     T cost;
88     T flow;
89 };
90
91 vector<Edge> edges() {
92     vector<Edge> a;
93     for (int i = 0; i < e.size(); i += 2) {
94         Edge x;
95         x.from = e[i + 1].to;
96         x.to = e[i].to;
97         x.cap = e[i].cap + e[i + 1].cap;
98         x.cost = e[i].cost;
99         x.flow = e[i + 1].cap;
100         a.push_back(x);
101     }
102     return a;
103 }
104 };

```

5 数论

5.1 扩展欧几里得算法

```

1 i64 exgcd(i64 a, i64 b, i64& x, i64& y) {
2     if (b == 0) {
3         x = 1, y = 0;
4         return a;
5     }
6     i64 d = exgcd(b, a % b, y, x);

```

```

7     y -= a / b * x;
8     return d;
9 }
10
11 i64 inv(i64 a, i64 m) {
12     i64 x, y;
13     i64 d = exgcd(a, m, x, y);
14     if (d != 1) {
15         return -1;
16     }
17     return (x % m + m) % m;
18 }
19
20 bool solveLinear(i64 a, i64 b, i64 c, i64& x, i64& y) {
21     i64 d = exgcd(a, b, x, y);
22     if (c % d != 0) {
23         return false;
24     }
25     i64 k = c / d;
26     x *= k;
27     y *= k;
28     return true;
29 }
30
31 pair<i64, i64> solveMod(i64 a, i64 b, i64 m) {
32     assert(m > 0);
33     b *= -1;
34     i64 x, y;
35     i64 g = exgcd(a, m, x, y);
36     if (g < 0) {
37         g *= -1;
38         x *= -1;
39         y *= -1;
40     }
41     if (b % g != 0) {
42         return {-1, -1};
43     }
44     x = x * (b / g) % (m / g);
45     if (x < 0) {
46         x += m / g;
47     }
48     return {x, m / g};
49 }

```

5.2 线性筛

```

1 vector<int> minp, primes;
2
3 void sieve(int n) {
4     minp.assign(n + 1, 0);
5     primes.clear();
6
7     for (int i = 2; i <= n; i++) {
8         if (minp[i] == 0) {
9             minp[i] = i;
10            primes.push_back(i);
11        }
12
13        for (auto p : primes) {
14            if (i * p > n) {
15                break;
16            }
17            minp[i * p] = p;
18            if (p == minp[i]) {
19                break;
20            }
21        }
22    }
23 }

```

5.3 莫比乌斯反演

莫比乌斯反演用来在「按约数求和」和「按倍数求和」之间互转。

$\mu(n)$ 的定义: $n = 1$ 时为 1; n 是 k 个互异素数之积时为 $(-1)^k$; 含平方因子时为 0。一切从这条恒等式出发:

$$\sum_{d|n} \mu(d) = [n = 1].$$

于是有两种方向的反演:

$$f(n) = \sum_{d|n} g(d) \implies g(n) = \sum_{d|n} \mu(d) f\left(\frac{n}{d}\right),$$

$$f(n) = \sum_{n|d} g(d) \implies g(n) = \sum_{n|d} \mu\left(\frac{d}{n}\right) f(d).$$

实际做题更常用的是「换条件」的用法: 把 $[\gcd(i, j) = 1]$ 换成 $\sum_{d|\gcd(i, j)} \mu(d)$, 再交换求和顺序, 条件就从 $\gcd$ 挪到了 d :

$$\sum_{i=1}^n \sum_{j=1}^m [\gcd(i, j) = 1] = \sum_{d=1}^{\min(n, m)} \mu(d) \left\lfloor \frac{n}{d} \right\rfloor \left\lfloor \frac{m}{d} \right\rfloor.$$

右边只要 μ 的前缀和就能整除分块, μ 用下面的 `sieve()` 线性筛出来。

```

1 vector<int> minp, primes, mu;
2
3 void sieve(int n) {
4     minp.assign(n + 1, 0);
5     mu.assign(n + 1, 0);
6     primes.clear();
7     mu[1] = 1;
8
9     for (int i = 2; i <= n; i++) {
10        if (minp[i] == 0) {
11            minp[i] = i;
12            mu[i] = -1;
13            primes.push_back(i);
14        }
15
16        for (auto p : primes) {
17            if (i * p > n) {
18                break;
19            }
20            minp[i * p] = p;
21            if (p == minp[i]) {
22                mu[i * p] = 0;
23                break;
24            } else {
25                mu[i * p] = -mu[i];
26            }
27        }
28    }
29 }
30

```

5.4 欧拉函数

$$\varphi(n) = n \prod_{i=1}^k \left(1 - \frac{1}{p_i}\right).$$

两条常用性质: 它是积性的, 即 $\gcd(a, b) = 1$ 时 $\varphi(ab) = \varphi(a)\varphi(b)$; 并且 $\sum_{d|n} \varphi(d) = n$, 也即

$$\varphi(n) = \sum_{d|n} \mu(d) \frac{n}{d}.$$

欧拉定理: $\gcd(a, m) = 1$ 时 $a^{\varphi(m)} \equiv 1 \pmod{m}$ 。竞赛里更常用的是扩展欧拉定理, 它给指数降幂—— $b \geq \varphi(m)$ 时对任意 a 都有

$$a^b \equiv a^{b \bmod \varphi(m) + \varphi(m)} \pmod{m}.$$

线性筛求 φ 的递推: 素数的 $\varphi(p) = p - 1$; 对 $i \cdot p$, 若 $p \mid i$ 则 $\varphi(ip) = p\varphi(i)$, 否则 $\varphi(ip) = (p - 1)\varphi(i)$ 。见下面的 `sieve()`; 只求单个数用 `Phi()`, 复杂度 $O(\sqrt{n})$ 。

```

1 vector<int> minp, primes, phi;
2
3 // 线性筛 (用于求解全部数的欧拉函数)
4 void sieve(int n) {
5     minp.assign(n + 1, 0);
6     phi.assign(n + 1, 0);
7     primes.clear();
8     phi[1] = 1;
9
10    for (int i = 2; i <= n; i++) {
11        if (minp[i] == 0) {
12            minp[i] = i;
13            primes.push_back(i);
14            phi[i] = i - 1;
15        }
16
17        for (auto p : primes) {
18            if (i * p > n) {
19                break;
20            }
21            minp[i * p] = p;
22            if (p == minp[i]) {
23                phi[i * p] = phi[i] * p;
24                break;
25            } else {
26                phi[i * p] = phi[i] * (p - 1);
27            }
28        }
29    }
30 }
31
32 // 求解单个数的欧拉函数
33 int Phi(int n) {
34     int res = n;
35     for (int i = 2; i * i <= n; i++) {
36         if (n % i == 0) {
37             while (n % i == 0) {
38                 n /= i;
39             }
40             res = res / i * (i - 1);
41         }
42     }
43     if (n > 1) {

```

```

44         res = res / n * (n - 1);
45     }
46     return res;
47 }
48

```

5.5 Miller-Rabin

```

1 bool isPrime(i64 n) {
2     if (n < 2) {
3         return false;
4     }
5     constexpr array<int, 10> smallPrimes = {2, 3, 5, 7, 11, 13, 17, 19, 23, 29};
6     for (auto& p : smallPrimes) {
7         if (n % p == 0) {
8             return n == p;
9         }
10    }
11    if (n < 31 * 31) {
12        return true;
13    }
14    constexpr array<int, 7> bases = {2, 325, 9375, 28178, 450775, 9780504,
15    1795265022};
16    int s = __builtin_ctzll(n - 1);
17    i64 d = (n - 1) >> s;
18    for (auto& a : bases) {
19        if (a % n == 0) {
20            continue;
21        }
22        i64 cur = power(a, d, n);
23        if (cur == 1) {
24            continue;
25        }
26        bool witness = true;
27        for (int r = 0; r < s; r++) {
28            if (cur == n - 1) {
29                witness = false;
30                break;
31            }
32            cur = mul(cur, cur, n);
33        }
34        if (witness) {
35            return false;
36        }
37    }
38    return true;
39 }

```

```

36 }
37 return true;
38 }

```

5.6 Pollard-Rho

```

1 static mt19937_64
  rng(static_cast<u32>(chrono::steady_clock::now().time_since_epoch().count()));
2
3 i64 rho(i64 n, i64 c) {
4     auto f = [&](i64 x) {
5         return static_cast<i64>((static_cast<i128>(x) * x + c) % n);
6     };
7     i64 x = 1, y = 2, z = 1, q = 1;
8     i64 g = 1;
9     constexpr i64 m = 127;
10    for (i64 r = 1; g == 1; r <= 1) {
11        x = y;
12        for (i64 i = 0; i < r; i++) {
13            y = f(y);
14        }
15        for (i64 k = 0; k < r && g == 1; k += m) {
16            z = y;
17            for (i64 i = 0; i < min(m, r - k); i++) {
18                y = f(y);
19                q = mul(q, abs(x - y), n);
20            }
21            g = gcd(q, n);
22        }
23    }
24    if (g == n) {
25        do {
26            z = f(z);
27            g = gcd(abs(x - z), n);
28        } while (g == 1);
29    }
30    return g;
31 }
32
33 i64 primeFactor(i64 n) {
34     assert(n >= 2);
35     if (n % 2 == 0) {
36         return 2;
37     }

```

```

38     while (!isPrime(n)) {
39         uniform_int_distribution<i64> pick(1, n - 1);
40         n = rho(n, pick(rng));
41     }
42     return n;
43 }
44
45 vector<pair<i64, int>> factorize(i64 n) {
46     vector<pair<i64, int>> pf;
47     for (int p = 2; p < 100; p++) {
48         if (p * p > n) {
49             break;
50         }
51         if (n % p == 0) {
52             int e = 0;
53             do {
54                 n /= p;
55                 e++;
56             } while (n % p == 0);
57             pf.emplace_back(p, e);
58         }
59     }
60     while (n > 1) {
61         i64 p = primeFactor(n);
62         int e = 0;
63         do {
64             n /= p;
65             e++;
66         } while (n % p == 0);
67         pf.emplace_back(p, e);
68     }
69     sort(pf.begin(), pf.end());
70     return pf;
71 }

```

5.7 约数构造

```

1 vector<i64> buildDivisors(const vector<pair<i64, int>>& factors) {
2     vector<i64> divisors = {1};
3     for (auto& [p, e] : factors) {
4         int sz = divisors.size();
5         for (int i = 0; i < sz; i++) {
6             i64 x = divisors[i];
7             for (int j = 0; j < e; j++) {

```

```

8         x *= p;
9         divisors.push_back(x);
10    }
11 }
12 }
13 return divisors;
14 }

```

6 组合数学

6.1 组合数

```

1 template <typename T>
2 struct Combination {
3     int n;
4     vector<T> fact_, invfact_, inv_;
5
6     Combination() : n(0), fact_{1}, invfact_{1}, inv_{0} {}
7     Combination(int n_) : Combination() {
8         init(n_);
9     }
10
11     void init(int n_) {
12         if (n_ <= n) {
13             return;
14         }
15         fact_.resize(n_ + 1);
16         invfact_.resize(n_ + 1);
17         inv_.resize(n_ + 1);
18
19         for (int i = n + 1; i <= n_; i++) {
20             fact_[i] = fact_[i - 1] * i;
21         }
22         invfact_[n_] = fact_[n_].inv();
23         for (int i = n_; i > n; i--) {
24             invfact_[i - 1] = invfact_[i] * i;
25             inv_[i] = invfact_[i] * fact_[i - 1];
26         }
27         n = n_;
28     }
29
30     T fact(int k) {
31         if (k > n) {
32             init(2 * k);

```

```

33     }
34     return fact_[k];
35 }
36
37 T invfact(int k) {
38     if (k > n) {
39         init(2 * k);
40     }
41     return invfact_[k];
42 }
43
44 T inv(int k) {
45     if (k > n) {
46         init(2 * k);
47     }
48     return inv_[k];
49 }
50
51 T binom(int n, int m) {
52     if (n < m || m < 0) {
53         return 0;
54     }
55     return fact(n) * invfact(m) * invfact(n - m);
56 }
57
58 T perm(int n, int m) {
59     if (n < m || m < 0) {
60         return 0;
61     }
62     return fact(n) * invfact(n - m);
63 }
64
65 T multicomcomb(int n, int m) {
66     if (n < 0 || m < 0) {
67         return 0;
68     }
69     if (n == 0) {
70         return (m == 0);
71     }
72     return binom(n + m - 1, m);
73 }
74
75 T Catalan(int n) {
76     if (n < 0) {

```

```

77     return 0;
78 }
79 return binom(2 * n, n) * inv(n + 1);
80 }
81 };

```

6.2 康托展开

```

1 vector<int> toLehmer(vector<int>& p) {
2     int n = p.size();
3     Fenwick<AddMono<int>> fen(n);
4     vector<int> c(n);
5     for (int i = n - 1; i >= 0; i--) {
6         c[i] = fen.sum(p[i]);
7         fen.add(p[i], 1);
8     }
9     return c;
10 }
11
12 vector<int> fromLehmer(vector<int>& c) {
13     int n = c.size();
14     vector<int> p(n);
15     Fenwick<AddMono<int>> fen(n);
16     for (int i = 0; i < n; i++) {
17         fen.add(i, 1);
18     }
19     for (int i = 0; i < n; i++) {
20         p[i] = fen.kth(c[i]);
21         fen.add(p[i], -1);
22     }
23     return p;
24 }
25
26 i64 toRank(vector<int>& p) {
27     int n = p.size();
28     auto c = toLehmer(p);
29     i64 fac = 1, rk = 0;
30     for (int i = n - 1; i >= 0; i--) {
31         rk += c[i] * fac;
32         fac *= (n - i);
33     }
34     return rk;
35 }
36

```

```

37 vector<int> fromRank(int n, i64 rk) {
38     vector<int> c(n);
39     for (int i = 1; i <= n; i++) {
40         c[n - i] = rk % i;
41         rk /= i;
42     }
43     return fromLehmer(c);
44 }

```

6.3 逆序对

```

1 i64 invCount(vector<i64>& a, int l, int r, vector<i64>& tmp) {
2     if (r - l <= 1) return 0;
3
4     i64 res = 0;
5     int mid = (l + r) / 2;
6     res += invCount(a, l, mid, tmp);
7     res += invCount(a, mid, r, tmp);
8
9     int i = l, j = mid, k = l;
10    while (i < mid && j < r) {
11        if (a[i] <= a[j]) {
12            tmp[k++] = a[i++];
13        } else {
14            tmp[k++] = a[j++];
15            res += mid - i;
16        }
17    }
18    while (i < mid) {
19        tmp[k++] = a[i++];
20    }
21    while (j < r) {
22        tmp[k++] = a[j++];
23    }
24
25    for (i = l; i < r; i++) {
26        a[i] = tmp[i];
27    }
28
29    return res;
30 }
31

```

6.4 容斥原理

```

1 // cap(S) 表示 S 集合的交集大小
2
3 // 交集形式
4 template <typename T, typename F>
5 T ie(int n, F&& cap) {
6     T ans{};
7     for (int S = 1; S < (1 << n); S++) {
8         T cur = cap(S);
9         if (__builtin_popcount(S) & 1) {
10             ans += cur;
11         } else {
12             ans -= cur;
13         }
14     }
15     return ans;
16 }
17
18 // 补集形式
19 template <typename T, typename F>
20 T ieComp(int n, F&& cap) {
21     T ans{};
22     for (int S = 0; S < (1 << n); S++) {
23         T cur = cap(S);
24         if (__builtin_popcount(S) & 1) {
25             ans -= cur;
26         } else {
27             ans += cur;
28         }
29     }
30     return ans;
31 }
32
33 // 恰好满足 k 个性质的元素数量
34 template <typename T, typename F>
35 vector<T> ieExact(int n, F&& cap) {
36     vector<T> N(n + 1, T{});
37     for (int S = 0; S < (1 << n); S++) {
38         int k = __builtin_popcount(S);
39         N[k] += cap(S);
40     }
41
42     vector<T> binom(n + 2, T{}), E(n + 1, T{});

```

```

43     binom[0] = T{1};
44     for (int j = 0; j <= n; j++) {
45         for (int k = 0; k <= j; k++) {
46             T term = N[j] * binom[k];
47             if ((j - k) & 1) {
48                 E[k] -= term;
49             } else {
50                 E[k] += term;
51             }
52         }
53         for (int k = j + 1; k >= 1; k--) {
54             binom[k] += binom[k - 1];
55         }
56     }
57     return E;
58 }
59
60 // 至少满足 k 个
61 template <typename T, typename F>
62 vector<T> ieLeast(int n, F&& cap) {
63     auto E = ieExact<T>(n, cap);
64     for (int k = n - 1; k >= 0; k--) {
65         E[k] += E[k + 1];
66     }
67     return E;
68 }
69
70 // 至多满足 k 个
71 template <typename T, typename F>
72 vector<T> ieMost(int n, F&& cap) {
73     auto E = ieExact<T>(n, cap);
74     for (int k = 0; k < n; k++) {
75         E[k + 1] += E[k];
76     }
77     return E;
78 }

```

6.5 zeta 变换

```

1 // 子集和: f[mask] = sum_{sub ⊆ mask} f[sub]
2 // 方向: 小的 mask → 大的 mask ("吸收"不带位)
3 template <typename T>
4 void subsetZeta(vector<T>& f) {

```

```

5   int n = f.size();
6   for (int i = 1; i < n; i <= 1) {
7       for (int mask = 0; mask < n; mask++) {
8           if (mask & i) {
9               f[mask] += f[mask ^ i];
10          }
11      }
12  }
13 }
14
15 // 子集和的逆
16 template <typename T>
17 void subsetMobius(vector<T>& f) {
18     int n = f.size();
19     for (int i = 1; i < n; i <= 1) {
20         for (int mask = 0; mask < n; mask++) {
21             if (mask & i) {
22                 f[mask] -= f[mask ^ i];
23             }
24         }
25     }
26 }
27
28 // 超集和: f[mask] = sum_{sup ⊇ mask} f[sup]
29 // 方向: 大的 mask → 小的 mask ("分发"给不带这一位的 mask)
30 template <typename T>
31 void supersetZeta(vector<T>& f) {
32     int n = f.size();
33     for (int i = 1; i < n; i <= 1) {
34         for (int mask = 0; mask < n; mask++) {
35             if (mask & i) {
36                 f[mask ^ i] += f[mask];
37             }
38         }
39     }
40 }
41
42 // 超集和的逆
43 template <typename T>
44 void supersetMobius(vector<T>& f) {
45     int n = f.size();
46     for (int i = 1; i < n; i <= 1) {
47         for (int mask = 0; mask < n; mask++) {

```

```

48         if (mask & i) {
49             f[mask ^ i] -= f[mask];
50         }
51     }
52 }
53 }

```

6.6 Polyomino

```

1 struct Polyomino {
2     using Cell = pair<int, int>;
3     set<Cell> cells;
4     Polyomino() {}
5     Polyomino(const set<Cell>& cells_) : cells(cells_) {}
6 };
7
8 Polyomino normalize(const Polyomino& p) {
9     if (p.cells.empty()) {
10         return p;
11     }
12     int minX = p.cells.begin()->first;
13     int minY = p.cells.begin()->second;
14     for (auto& [x, y] : p.cells) {
15         minX = min(minX, x);
16         minY = min(minY, y);
17     }
18     Polyomino res;
19     for (auto& [x, y] : p.cells) {
20         res.cells.emplace(x - minX, y - minY);
21     }
22     return res;
23 }
24
25 Polyomino rotate(const Polyomino& p) {
26     Polyomino res;
27     for (auto& [x, y] : p.cells) {
28         res.cells.emplace(y, -x);
29     }
30     return normalize(res);
31 }
32
33 Polyomino flip(const Polyomino& p) {
34     Polyomino res;
35     for (auto& [x, y] : p.cells) {

```

```

36     res.cells.emplace(x, -y);
37 }
38 return normalize(res);
39 }

```

7 数据结构

7.1 线段树

7.1.1 线段树

```

1 template <typename Mono>
2 struct SegmentTree {
3     using X = typename Mono::Type;
4
5     int n;
6     vector<X> info;
7
8     SegmentTree() : n(0) {}
9     SegmentTree(const vector<X>& init_) {
10         init(init_);
11     }
12
13     void init(const vector<X>& init_) {
14         n = init_.size();
15         int logn = __lg(n);
16         info.assign(4 << logn, Mono::e());
17
18         auto build = [&](auto&& self, int p, int l, int r) -> void {
19             if (r - l == 1) {
20                 info[p] = init_[l];
21                 return;
22             }
23             int mid = (l + r) / 2;
24             self(self, 2 * p, l, mid);
25             self(self, 2 * p + 1, mid, r);
26             pull(p);
27         };
28
29         build(build, 1, 0, n);
30     }
31
32     void pull(int p) {
33         info[p] = Mono::op(info[2 * p], info[2 * p + 1]);
34     }

```

```

35
36     void modify(int p, int l, int r, int x, const X& v) {
37         if (r - l == 1) {
38             info[p] = v;
39             return;
40         }
41         int mid = (l + r) / 2;
42         if (x < mid) {
43             modify(2 * p, l, mid, x, v);
44         } else {
45             modify(2 * p + 1, mid, r, x, v);
46         }
47         pull(p);
48     }
49     void modify(int p, const X& v) {
50         modify(1, 0, n, p, v);
51     }
52
53     X rangeQuery(int p, int l, int r, int x, int y) {
54         if (y <= l || r <= x) {
55             return Mono::e();
56         }
57         if (x <= l && r <= y) {
58             return info[p];
59         }
60         int mid = (l + r) / 2;
61         return Mono::op(rangeQuery(2 * p, l, mid, x, y), rangeQuery(2 * p + 1, mid,
62 r, x, y));
63     }
64     X rangeQuery(int l, int r) {
65         return rangeQuery(1, 0, n, l, r);
66     }
67
68     template <typename F>
69     int findFirst(int p, int l, int r, int x, int y, F&& pred) {
70         if (y <= l || r <= x) {
71             return -1;
72         }
73         if (x <= l && r <= y && !pred(info[p])) {
74             return -1;
75         }
76         if (r - l == 1) {
77             return l;

```

```

77     }
78     int mid = (l + r) / 2;
79     int res = findFirst(2 * p, l, mid, x, y, pred);
80     if (res == -1) {
81         res = findFirst(2 * p + 1, mid, r, x, y, pred);
82     }
83     return res;
84 }
85 template <typename F>
86 int findFirst(int l, int r, F&& pred) {
87     return findFirst(1, 0, n, l, r, pred);
88 }
89
90 template <typename F>
91 int findLast(int p, int l, int r, int x, int y, F&& pred) {
92     if (y <= l || r <= x) {
93         return -1;
94     }
95     if (x <= l && r <= y && !pred(info[p])) {
96         return -1;
97     }
98     if (r - l == 1) {
99         return l;
100     }
101     int mid = (l + r) / 2;
102     int res = findLast(2 * p + 1, mid, r, x, y, pred);
103     if (res == -1) {
104         res = findLast(2 * p, l, mid, x, y, pred);
105     }
106     return res;
107 }
108 template <typename F>
109 int findLast(int l, int r, F&& pred) {
110     return findLast(1, 0, n, l, r, pred);
111 }
112 };

```

7.1.2 懒标记线段树

```

1 template <typename ActMono>
2 struct LazySegmentTree {
3     using Mono = typename ActMono::Mono;
4     using Tag = typename ActMono::Tag;
5     using X = typename ActMono::X;

```

```

6     using Y = typename ActMono::Y;
7
8     int n;
9     vector<X> info;
10    vector<Y> tag;
11
12    LazySegmentTree() : n(0) {}
13    LazySegmentTree(const vector<X>& init_) {
14        init(init_);
15    }
16
17    void init(const vector<X>& init_) {
18        n = init_.size();
19        int logn = __lg(n);
20        info.assign(4 << logn, Mono::e());
21        tag.assign(4 << logn, Tag::e());
22
23        auto build = [&](auto&& self, int p, int l, int r) -> void {
24            if (r - l == 1) {
25                info[p] = init_[l];
26                return;
27            }
28            int mid = (l + r) / 2;
29            self(self, 2 * p, l, mid);
30            self(self, 2 * p + 1, mid, r);
31            pull(p);
32        };
33
34        build(build, 1, 0, n);
35    }
36
37    void pull(int p) {
38        info[p] = Mono::op(info[2 * p], info[2 * p + 1]);
39    }
40
41    void apply(int p, int len, const Y& v) {
42        info[p] = ActMono::act(info[p], v, len);
43        tag[p] = Tag::op(tag[p], v);
44    }
45
46    void push(int p, int len) {
47        int llen = len / 2;
48        apply(2 * p, llen, tag[p]);
49        apply(2 * p + 1, len - llen, tag[p]);

```

```

50     tag[p] = Tag::e();
51 }
52
53 void modify(int p, int l, int r, int x, const X& v) {
54     if (r - l == 1) {
55         info[p] = v;
56         return;
57     }
58     int mid = (l + r) / 2;
59     push(p, r - l);
60     if (x < mid) {
61         modify(2 * p, l, mid, x, v);
62     } else {
63         modify(2 * p + 1, mid, r, x, v);
64     }
65     pull(p);
66 }
67 void modify(int p, const X& v) {
68     modify(1, 0, n, p, v);
69 }
70
71 X rangeQuery(int p, int l, int r, int x, int y) {
72     if (y <= l || r <= x) {
73         return Mono::e();
74     }
75     if (x <= l && r <= y) {
76         return info[p];
77     }
78     int mid = (l + r) / 2;
79     push(p, r - l);
80     return Mono::op(rangeQuery(2 * p, l, mid, x, y), rangeQuery(2 * p + 1, mid,
81 r, x, y));
82 }
83 X rangeQuery(int l, int r) {
84     return rangeQuery(1, 0, n, l, r);
85 }
86
87 void rangeApply(int p, int l, int r, int x, int y, const Y& v) {
88     if (y <= l || r <= x) {
89         return;
90     }
91     if (x <= l && r <= y) {
92         apply(p, r - l, v);

```

```

92         return;
93     }
94     int mid = (l + r) / 2;
95     push(p, r - l);
96     rangeApply(2 * p, l, mid, x, y, v);
97     rangeApply(2 * p + 1, mid, r, x, y, v);
98     pull(p);
99 }
100 void rangeApply(int l, int r, const Y& v) {
101     return rangeApply(1, 0, n, l, r, v);
102 }
103
104 template <typename F>
105 int findFirst(int p, int l, int r, int x, int y, F&& pred) {
106     if (y <= l || r <= x) {
107         return -1;
108     }
109     if (x <= l && r <= y && !pred(info[p])) {
110         return -1;
111     }
112     if (r - l == 1) {
113         return l;
114     }
115     int mid = (l + r) / 2;
116     push(p, r - l);
117     int res = findFirst(2 * p, l, mid, x, y, pred);
118     if (res == -1) {
119         res = findFirst(2 * p + 1, mid, r, x, y, pred);
120     }
121     return res;
122 }
123 template <typename F>
124 int findFirst(int l, int r, F&& pred) {
125     return findFirst(1, 0, n, l, r, pred);
126 }
127
128 template <typename F>
129 int findLast(int p, int l, int r, int x, int y, F&& pred) {
130     if (y <= l || r <= x) {
131         return -1;
132     }
133     if (x <= l && r <= y && !pred(info[p])) {
134         return -1;

```

```

135     }
136     if (r - l == 1) {
137         return l;
138     }
139     int mid = (l + r) / 2;
140     push(p, r - l);
141     int res = findLast(2 * p + 1, mid, r, x, y, pred);
142     if (res == -1) {
143         res = findLast(2 * p, l, mid, x, y, pred);
144     }
145     return res;
146 }
147 template <typename F>
148 int findLast(int l, int r, F&& pred) {
149     return findLast(1, 0, n, l, r, pred);
150 }
151 };

```

7.1.3 动态开点线段树

```

1 template <typename ActMono>
2 struct DynamicSegmentTree {
3     using Mono = typename ActMono::Mono;
4     using Tag = typename ActMono::Tag;
5     using X = typename ActMono::X;
6     using Y = typename ActMono::Y;
7
8     int n;
9     int root;
10    vector<X> info;
11    vector<Y> tag;
12    vector<int> lc, rc;
13
14    DynamicSegmentTree() : n(0), root(-1) {}
15    DynamicSegmentTree(int n_) {
16        init(n_);
17    }
18
19    void init(int n_) {
20        n = n_;
21        info.clear();
22        tag.clear();
23        lc.clear();
24        rc.clear();

```

```

25        root = newNode();
26    }
27
28    int newNode() {
29        info.push_back(Mono::e());
30        tag.push_back(Tag::e());
31        lc.push_back(-1);
32        rc.push_back(-1);
33        return int(info.size()) - 1;
34    }
35
36    void pull(int p) {
37        X left = (lc[p] != -1 ? info[lc[p]] : Mono::e());
38        X right = (rc[p] != -1 ? info[rc[p]] : Mono::e());
39        info[p] = Mono::op(left, right);
40    }
41
42    void apply(int p, int len, const Y& v) {
43        info[p] = ActMono::act(info[p], v, len);
44        tag[p] = Tag::op(tag[p], v);
45    }
46
47    void push(int p, int len) {
48        if (lc[p] == -1) {
49            lc[p] = newNode();
50        }
51        if (rc[p] == -1) {
52            rc[p] = newNode();
53        }
54        int llen = len / 2;
55        apply(lc[p], llen, tag[p]);
56        apply(rc[p], len - llen, tag[p]);
57        tag[p] = Tag::e();
58    }
59
60    void modify(int p, int l, int r, int x, const X& v) {
61        if (r - l == 1) {
62            info[p] = v;
63            return;
64        }
65        int mid = (l + r) / 2;
66        push(p, r - l);
67        if (x < mid) {

```

```

68     modify(lc[p], l, mid, x, v);
69 } else {
70     modify(rc[p], mid, r, x, v);
71 }
72 pull(p);
73 }
74 void modify(int x, const X& v) {
75     modify(root, 0, n, x, v);
76 }
77
78 X rangeQuery(int p, int l, int r, int x, int y) {
79     if (p == -1 || y <= l || r <= x) {
80         return Mono::e();
81     }
82     if (x <= l && r <= y) {
83         return info[p];
84     }
85     int mid = (l + r) / 2;
86     push(p, r - l);
87     return Mono::op(rangeQuery(lc[p], l, mid, x, y), rangeQuery(rc[p], mid, r,
x, y));
88 }
89 X rangeQuery(int l, int r) {
90     return rangeQuery(root, 0, n, l, r);
91 }
92
93 void rangeApply(int p, int l, int r, int x, int y, const Y& v) {
94     if (y <= l || r <= x) {
95         return;
96     }
97     if (x <= l && r <= y) {
98         apply(p, r - l, v);
99         return;
100     }
101     int mid = (l + r) / 2;
102     push(p, r - l);
103     rangeApply(lc[p], l, mid, x, y, v);
104     rangeApply(rc[p], mid, r, x, y, v);
105     pull(p);
106 }
107 void rangeApply(int l, int r, const Y& v) {
108     rangeApply(root, 0, n, l, r, v);
109 }

```

```

110
111 template <typename F>
112 int findFirst(int p, int l, int r, int x, int y, F&& pred) {
113     if (p == -1 || y <= l || r <= x) {
114         return -1;
115     }
116     if (x <= l && r <= y && !pred(info[p])) {
117         return -1;
118     }
119     if (r - l == 1) {
120         return l;
121     }
122     int mid = (l + r) / 2;
123     push(p, r - l);
124     int res = findFirst(lc[p], l, mid, x, y, pred);
125     if (res == -1) {
126         res = findFirst(rc[p], mid, r, x, y, pred);
127     }
128     return res;
129 }
130
131 template <typename F>
132 int findFirst(int l, int r, F&& pred) {
133     return findFirst(root, 0, n, l, r, pred);
134 }
135
136 template <typename F>
137 int findLast(int p, int l, int r, int x, int y, F&& pred) {
138     if (p == -1 || y <= l || r <= x) {
139         return -1;
140     }
141     if (x <= l && r <= y && !pred(info[p])) {
142         return -1;
143     }
144     if (r - l == 1) {
145         return l;
146     }
147     int mid = (l + r) / 2;
148     push(p, r - l);
149     int res = findLast(rc[p], mid, r, x, y, pred);
150     if (res == -1) {
151         res = findLast(lc[p], l, mid, x, y, pred);
152     }
153     return res;

```

```

153 }
154 template <typename F>
155 int findLast(int l, int r, F&& pred) {
156     return findLast(root, 0, n, l, r, pred);
157 }
158 };

```

7.2 树状数组

7.2.1 树状数组

```

1 template <typename Mono>
2 struct Fenwick {
3     using X = typename Mono::Type;
4
5     int n;
6     vector<X> a;
7
8     Fenwick() : n(0) {}
9     Fenwick(int n_) {
10         init(n_);
11     }
12
13     void init(int n_) {
14         n = n_;
15         a.assign(n, Mono::e());
16     }
17
18     void add(int x, const X& v) {
19         assert(0 <= x && x < n);
20         for (int i = x + 1; i <= n; i += i & -i) {
21             a[i - 1] = Mono::op(a[i - 1], v);
22         }
23     }
24
25     X sum(int x) {
26         assert(0 <= x && x <= n);
27         X res = Mono::e();
28         for (int i = x; i > 0; i -= i & -i) {
29             res = Mono::op(a[i - 1], res);
30         }
31         return res;
32     }
33
34     X rangeSum(int l, int r) {

```

```

35         assert(l <= r);
36         return Mono::op(Mono::inv(sum(l)), sum(r));
37     }
38
39     // lower_bound
40     int kth(X k) {
41         int x = 0;
42         for (int i = 1 << __lg(n); i; i /= 2) {
43             if (x + i <= n && k >= a[x + i - 1]) {
44                 x += i;
45                 k -= a[x - 1];
46             }
47         }
48         return x;
49     }
50 };

```

7.2.2 区间加、单点查树状数组

```

1 template <typename Abel>
2 struct DualFenwick {
3     using X = typename Abel::Type;
4
5     int n;
6     Fenwick<Abel> fen;
7
8     DualFenwick() : n(0), fen(0) {}
9     DualFenwick(int n_) {
10         init(n_);
11     }
12
13     void init(int n_) {
14         n = n_;
15         fen.init(n);
16     }
17
18     void add(int l, int r, const X& v) {
19         assert(0 <= l && l <= r && r <= n);
20         fen.add(l, v);
21         if (r < n) {
22             fen.add(r, Abel::inv(v));
23         }
24     }
25
26     X query(int x) {

```

```

27     assert(0 <= x && x < n);
28     return fen.sum(x + 1);
29 }
30 };

```

7.2.3 区间加、区间和树状数组

```

1 template <typename Abel>
2 struct FenwickRangeAdd {
3     using X = typename Abel::Type;
4
5     int n;
6     Fenwick<Abel> fen0, fen1;
7
8     FenwickRangeAdd() : n(0), fen0(1), fen1(1) {}
9     FenwickRangeAdd(int n_) {
10         init(n_);
11     }
12
13     void init(int n_) {
14         n = n_;
15         fen0.init(n + 1);
16         fen1.init(n + 1);
17     }
18
19     void add(int l, int r, const X& v) {
20         assert(0 <= l && l <= r && r <= n);
21         fen0.add(l, Abel::power(v, -static_cast<i64>(l)));
22         fen0.add(r, Abel::power(v, static_cast<i64>(r)));
23         fen1.add(l, v);
24         fen1.add(r, Abel::inv(v));
25     }
26
27     X sum(int r) {
28         assert(0 <= r && r <= n);
29         return Abel::op(Abel::power(fen1.sum(r), static_cast<i64>(r)),
30 fen0.sum(r));
31     }
32
33     X rangeSum(int l, int r) {
34         assert(0 <= l && l <= r && r <= n);
35         return Abel::op(Abel::inv(sum(l)), sum(r));
36 };

```

7.2.4 二维树状数组

```

1 template <typename Mono>
2 struct Fenwick2D {
3     using X = typename Mono::Type;
4
5     int h, w;
6     vector<X> a;
7
8     Fenwick2D() : h(0), w(0) {}
9     Fenwick2D(int h_, int w_) {
10         init(h_, w_);
11     }
12
13     void init(int h_, int w_) {
14         h = h_;
15         w = w_;
16         a.assign((h + 1) * (w + 1), Mono::e());
17     }
18
19     void add(int x, int y, const X& v) {
20         assert(0 <= x && x < h);
21         assert(0 <= y && y < w);
22         for (int i = x + 1; i <= h; i += i & -i) {
23             for (int j = y + 1; j <= w; j += j & -j) {
24                 a[i * (w + 1) + j] = Mono::op(a[i * (w + 1) + j], v);
25             }
26         }
27     }
28
29     X sum(int x, int y) {
30         assert(0 <= x && x <= h);
31         assert(0 <= y && y <= w);
32         X res = Mono::e();
33         for (int i = x; i > 0; i -= i & -i) {
34             for (int j = y; j > 0; j -= j & -j) {
35                 res = Mono::op(res, a[i * (w + 1) + j]);
36             }
37         }
38         return res;
39     }
40
41     X rangeSum(int x1, int x2, int y1, int y2) {
42         assert(0 <= x1 && x1 <= x2 && x2 <= h);
43         assert(0 <= y1 && y1 <= y2 && y2 <= w);

```

```

44     X s22 = sum(x2, y2);
45     X s12 = sum(x1, y2);
46     X s21 = sum(x2, y1);
47     X s11 = sum(x1, y1);
48     return Mono::op(Mono::op(s22, Mono::inv(s12)), Mono::op(Mono::inv(s21),
49 s11));
50 }
};

```

7.3 倍增

```

1 template <typename Mono>
2 struct BinaryLifting {
3     using X = typename Mono::Type;
4
5     int n, logn;
6     vector<vector<int>> nxt;
7     vector<vector<X>> info;
8
9     BinaryLifting() : n(0) {}
10    BinaryLifting(int n_) {
11        init(n_);
12    }
13
14    void init(int n_) {
15        n = n_;
16        logn = __lg(n);
17        nxt.assign(logn + 1, vector<int>(n, -1));
18        info.assign(logn + 1, vector<X>(n, Mono::e()));
19    }
20
21    void add(int u, int v, const X& x) {
22        assert(-1 <= v && v < n);
23        nxt[0][u] = v;
24        info[0][u] = x;
25    }
26
27    void build() {
28        for (int k = 0; k < logn; k++) {
29            for (int u = 0; u < n; u++) {
30                int v = nxt[k][u];
31                if (v == -1) {
32                    nxt[k + 1][u] = -1;
33                    info[k + 1][u] = info[k][u];

```

```

34                } else {
35                    nxt[k + 1][u] = nxt[k][v];
36                    info[k + 1][u] = Mono::op(info[k][u], info[k][v]);
37                }
38            }
39        }
40    }
41
42    pair<int, X> query(int u, i64 step) {
43        assert(0 <= step && step < (1LL << (logn + 1)));
44        X res = Mono::e();
45        for (int k = logn; k >= 0 && u != -1; k--) {
46            if ((step >> k) & 1) {
47                res = Mono::op(res, info[k][u]);
48                u = nxt[k][u];
49            }
50        }
51        return {u, res};
52    }
53
54    int jump(int u, i64 step) {
55        assert(0 <= step && step < (1LL << (logn + 1)));
56        for (int k = logn; k >= 0 && u != -1; k--) {
57            if ((step >> k) & 1) {
58                u = nxt[k][u];
59            }
60        }
61        return u;
62    }
63 };

```

7.4 并查集

```

1 struct DSU {
2     vector<int> f, siz;
3
4     DSU() {}
5     DSU(int n) {
6         init(n);
7     }
8
9     void init(int n) {
10        f.resize(n);
11        iota(f.begin(), f.end(), 0);

```

```

12     siz.assign(n, 1);
13 }
14
15 int find(int x) {
16     while (x != f[x]) {
17         x = f[x] = f[f[x]];
18     }
19     return x;
20 }
21
22 bool same(int x, int y) {
23     return find(x) == find(y);
24 }
25
26 bool merge(int x, int y) {
27     x = find(x);
28     y = find(y);
29     if (x == y) {
30         return false;
31     }
32     siz[x] += siz[y];
33     f[y] = x;
34     return true;
35 }
36
37 int size(int x) {
38     return siz[find(x)];
39 }
40 };

```

7.5 稀疏表

```

1 template <typename Mono>
2 struct SparseTable {
3     using X = typename Mono::Type;
4
5     int n;
6     vector<vector<X>> st;
7
8     SparseTable() : n(0) {}
9     SparseTable(const vector<X>& a) {
10         init(a);
11     }
12
13     void init(const vector<X>& a) {

```

```

14         n = a.size();
15         int logn = __lg(n);
16         st.assign(logn + 1, vector<X>(n));
17         for (int i = 0; i < n; i++) {
18             st[0][i] = a[i];
19         }
20         for (int j = 1; (1 << j) <= n; j++) {
21             for (int i = 0; i + (1 << j) <= n; i++) {
22                 st[j][i] = Mono::op(st[j - 1][i], st[j - 1][i + (1 << (j - 1))]);
23             }
24         }
25     }
26
27     X query(int l, int r) {
28         assert(l < r);
29         int len = r - l;
30         int t = __lg(len);
31         return Mono::op(st[t][l], st[t][r - (1 << t)]);
32     }
33 };

```

7.6 01-Trie

```

1 struct BinaryTrie {
2     struct Node {
3         array<int, 2> ch;
4         int end;
5         Node() {
6             ch[0] = ch[1] = 0;
7             end = 0;
8         }
9     };
10
11     vector<Node> tr;
12     int tot;
13
14     BinaryTrie(int n) {
15         init(n);
16     }
17
18     void init(int n) {
19         tr.resize(n);
20         tot = 1;
21     }

```

```

22
23 void insert(int x) {
24     int p = 1;
25     for (int i = 30; i >= 0; i--) {
26         int c = x >> i & 1;
27         if (!tr[p].ch[c]) {
28             ++tot;
29             tr[p].ch[c] = tot;
30         }
31         p = tr[p].ch[c];
32         tr[p].end++;
33     }
34 }
35
36 int query(int x) {
37     int p = 1, res = 0;
38     for (int i = 30; i >= 0; i--) {
39         int c = x >> i & 1;
40         if (tr[p].ch[c ^ 1]) {
41             res |= (1 << i);
42             p = tr[p].ch[c ^ 1];
43         } else {
44             p = tr[p].ch[c];
45         }
46     }
47     return res;
48 }
49
50 void erase(int x) {
51     int p = 1;
52     for (int i = 30; i >= 0; i--) {
53         int c = x >> i & 1;
54         int nxt = tr[p].ch[c];
55         tr[nxt].end--;
56         if (tr[nxt].end == 0) {
57             tr[p].ch[c] = 0;
58         }
59         p = nxt;
60     }
61 }
62 };
63

```

7.7 笛卡尔树

```

1 template <typename T, bool IS_MIN = false>
2 struct CartesianTree {
3     int n;
4     vector<T>& a;
5     vector<int> L, R, p;
6     // range[i]: the interval of the subtree rooted at i
7     vector<pair<int, int>> range;
8     int root;
9
10    CartesianTree(vector<T>& a_) : n(a_.size()), a(a_), root(-1) {
11        L.assign(n, -1);
12        R.assign(n, -1);
13        p.assign(n, -1);
14        range.assign(n, {0, n});
15        vector<int> stk;
16
17        auto cmp = [&](int i, int j) -> bool {
18            if (a[i] == a[j]) {
19                return i < j;
20            }
21            if constexpr (IS_MIN) {
22                return a[i] < a[j];
23            } else {
24                return a[j] < a[i];
25            }
26        };
27
28        for (int i = 0; i < n; i++) {
29            while (!stk.empty() && cmp(i, stk.back())) {
30                L[i] = stk.back();
31                stk.pop_back();
32            }
33            range[i].first = stk.empty() ? 0 : stk.back() + 1;
34            if (!stk.empty()) {
35                R[stk.back()] = i;
36            }
37            stk.push_back(i);
38        }
39        if (!stk.empty()) {
40            root = stk.front();
41        }
42        stk.clear();

```

```

43     for (int i = n - 1; i >= 0; i--) {
44         while (!stk.empty() && cmp(i, stk.back())) {
45             stk.pop_back();
46         }
47         range[i].second = stk.empty() ? n : stk.back();
48         stk.push_back(i);
49     }
50     for (int i = 0; i < n; i++) {
51         if (L[i] != -1) {
52             p[L[i]] = i;
53         }
54         if (R[i] != -1) {
55             p[R[i]] = i;
56         }
57     }
58 }
59 };

```

8 线性代数

8.1 矩阵

```

1  template <typename T>
2  struct Matrix {
3      int n, m;
4      vector<vector<T>>> a;
5
6      Matrix() : n(0), m(0) {}
7      Matrix(int n_, int m_, T v = T{}) : n(n_), m(m_), a(n, vector<T>(m, v)) {}
8      Matrix(int n_) : Matrix(n_, n_) {}
9
10     static Matrix identity(int n) {
11         Matrix res(n);
12         for (int i = 0; i < n; i++) {
13             res[i][i] = T{1};
14         }
15         return res;
16     }
17
18     vector<T>& operator[](int i) {
19         return a[i];
20     }
21     const vector<T>& operator[](int i) const {
22         return a[i];

```

```

23     }
24 };
25
26 template <typename T>
27 Matrix<T> operator*(const Matrix<T>& a, const Matrix<T>& b) {
28     assert(a.m == b.n);
29     Matrix<T> c(a.n, b.m);
30     for (int i = 0; i < c.n; i++) {
31         for (int k = 0; k < a.m; k++) {
32             for (int j = 0; j < c.m; j++) {
33                 c.a[i][j] += a.a[i][k] * b.a[k][j];
34             }
35         }
36     }
37     return c;
38 }
39
40 template <typename T>
41 Matrix<T>& operator*=(Matrix<T>& a, const Matrix<T>& b) {
42     return a = a * b;
43 }
44
45 template <typename T, integral U>
46 Matrix<T> power(Matrix<T> a, U b) {
47     Matrix<T> res = Matrix<T>::identity(a.n);
48     for (; b > 0; b >>= 1) {
49         if (b & 1) {
50             res *= a;
51         }
52         a *= a;
53     }
54     return res;
55 }
56
57 template <typename T>
58 Matrix<T> transpose(const Matrix<T>& a) {
59     Matrix<T> res(a.m, a.n);
60     for (int i = 0; i < a.n; i++) {
61         for (int j = 0; j < a.m; j++) {
62             res[j][i] = a[i][j];
63         }
64     }
65     return res;

```

```

66 }
67
68 template <typename T>
69 Matrix<T> augment(const Matrix<T>& a, const Matrix<T>& b) {
70     assert(a.n == b.n);
71     Matrix<T> res(a.n, a.m + b.m);
72     for (int i = 0; i < a.n; i++) {
73         for (int j = 0; j < a.m; j++) {
74             res[i][j] = a[i][j];
75         }
76         for (int j = 0; j < b.m; j++) {
77             res[i][a.m + j] = b[i][j];
78         }
79     }
80     return res;
81 }
82
83 template <typename T>
84 Matrix<T> augment(const Matrix<T>& a, const vector<T>& b) {
85     assert(a.n == int(b.size()));
86     Matrix<T> res(a.n, a.m + 1);
87     for (int i = 0; i < a.n; i++) {
88         for (int j = 0; j < a.m; j++) {
89             res[i][j] = a[i][j];
90         }
91         res[i][a.m] = b[i];
92     }
93     return res;
94 }

```

8.2 高斯消元

```

1 template <typename T>
2 pair<vector<int>, T> eliminate(Matrix<T>& A, int lim, bool full) {
3     // full == true -> RREF
4     // full == false -> REF
5     vector<int> pivot;
6     T prod = T{1};
7     int r = 0;
8     for (int j = 0; j < lim && r < A.n; j++) {
9         int p = r;
10        while (p < A.n && A[p][j] == T{0}) {
11            p++;
12        }

```

```

13         if (p == A.n) {
14             continue;
15         }
16         if (p != r) {
17             swap(A[p], A[r]);
18             prod = -prod;
19         }
20         T x = A[r][j];
21         prod *= x;
22         pivot.push_back(j);
23         T inv = T{1} / x;
24         for (int k = j; k < A.m; k++) {
25             A[r][k] *= inv;
26         }
27         for (int i = 0; i < A.n; i++) {
28             if (i == r || (!full && i < r)) {
29                 continue;
30             }
31             T c = A[i][j];
32             if (c == T{0}) {
33                 continue;
34             }
35             A[i][j] = T{0};
36             for (int k = j + 1; k < A.m; k++) {
37                 A[i][k] -= c * A[r][k];
38             }
39         }
40         r++;
41     }
42     return {pivot, prod};
43 }
44
45 template <typename T>
46 pair<vector<int>, T> gauss(Matrix<T>& A) {
47     return eliminate(A, A.m, false);
48 }
49
50 template <typename T>
51 pair<vector<int>, T> gaussJordan(Matrix<T>& A, int lim) {
52     return eliminate(A, lim, true);
53 }
54
55 template <typename T>
56 pair<T, Matrix<T>> matrixInv(const Matrix<T>& a) {

```

```

56     if (a.n != a.m) {
57         return {T{0}, {}};
58     }
59     int n = a.n;
60     auto aug = augment(a, Matrix<T>::identity(n));
61     auto [pivot, det] = gaussJordan(aug, n);
62     if (pivot.size() < n) {
63         return {T{0}, {}};
64     }
65     Matrix<T> inv(n);
66     for (int i = 0; i < n; i++) {
67         for (int j = 0; j < n; j++) {
68             inv[i][j] = aug[i][n + j];
69         }
70     }
71     return {det, inv};
72 }
73
74 template <typename T>
75 int matrixRank(Matrix<T> a) {
76     auto [pivot, prod] = gauss(a);
77     return pivot.size();
78 }
79
80 template <typename T>
81 T determinant(Matrix<T> A) {
82     assert(A.n == A.m);
83     auto [pivot, det] = gauss(A);
84     if (pivot.size() < A.n) {
85         return T{0};
86     }
87     return det;
88 }
89
90 template <typename T>
91 vector<vector<T>> solveLinear(const Matrix<T>& A, const vector<T>& b) {
92     int n = A.n;
93     int m = A.m;
94     auto aug = augment(A, b);
95     auto [pivot, prod] = gaussJordan(aug, m);
96     int r = pivot.size();
97     for (int i = r; i < n; i++) {
98         if (aug[i][m] != T{0}) {

```

```

99         return {};
100     }
101 }
102 vector<int> row(m, -1);
103 for (int i = 0; i < r; i++) {
104     row[pivot[i]] = i;
105 }
106 vector<vector<T>> sol(1, vector<T>(m));
107 for (int j = 0; j < m; j++) {
108     if (row[j] != -1) {
109         sol[0][j] = aug[row[j]][m];
110     }
111 }
112 for (int j = 0; j < m; j++) {
113     if (row[j] != -1) {
114         continue;
115     }
116     vector<T> x(m);
117     x[j] = T{-1};
118     for (int i = 0; i < r; i++) {
119         x[pivot[i]] = aug[i][j];
120     }
121     sol.push_back(x);
122 }
123 return sol;
124 }

```

8.3 GF(2) 空间

8.3.1 异或线性基

```

1 struct Basis {
2     array<u64, 64> p{};
3     bool norm = false;
4
5     int size() {
6         int r = 0;
7         for (int i = 0; i < 64; i++) {
8             r += p[i] != 0;
9         }
10        return r;
11    }
12
13    bool insert(u64 x) {

```

```

14     for (int i = 63; i >= 0; i--) {
15         if (!(x >> i & 1)) {
16             continue;
17         }
18         if (!p[i]) {
19             p[i] = x;
20             norm = false;
21             return true;
22         }
23         x ^= p[i];
24     }
25     return false;
26 }
27
28 bool contain(u64 x) {
29     for (int i = 63; i >= 0; i--) {
30         if (x >> i & 1) {
31             x ^= p[i];
32         }
33     }
34     return x == 0;
35 }
36
37 u64 getMax(u64 x = 0) {
38     for (int i = 63; i >= 0; i--) {
39         if ((x ^ p[i]) > x) {
40             x ^= p[i];
41         }
42     }
43     return x;
44 }
45
46 u64 getMin(u64 x = 0) {
47     for (int i = 63; i >= 0; i--) {
48         if ((x ^ p[i]) < x) {
49             x ^= p[i];
50         }
51     }
52     return x;
53 }
54
55 void normalize() {
56     if (norm == true) {

```

```

57         return;
58     }
59     norm = true;
60     for (int i = 63; i >= 0; i--) {
61         if (!p[i]) {
62             continue;
63         }
64         for (int j = i - 1; j >= 0; j--) {
65             if (p[i] >> j & 1) {
66                 p[i] ^= p[j];
67             }
68         }
69     }
70 }
71
72 u64 kth(u64 k) {
73     normalize();
74     vector<u64> b;
75     for (int i = 0; i < 64; i++) {
76         if (p[i]) {
77             b.push_back(p[i]);
78         }
79     }
80     int r = b.size();
81     if (r < 64) {
82         if (k < (1LL << r)) {
83             return -1;
84         }
85     }
86     u64 res = 0;
87     for (int i = 0; i < r; i++) {
88         if (k >> i & 1) {
89             res ^= b[i];
90         }
91     }
92     return res;
93 }
94 };

```

8.3.2 逆矩阵

```

1 namespace gf2 {
2
3 template <typename U>

```

```

4 vector<U> inverse(int n, vector<U> a) {
5     vector<U> b(n);
6     for (int i = 0; i < n; i++) {
7         b[i] = U{1} << i;
8     }
9     for (int j = 0; j < n; j++) {
10        int p = j;
11        while (p < n && !(a[p] >> j & U{1})) {
12            p++;
13        }
14        if (p == n) {
15            return {};
16        }
17        swap(a[p], a[j]);
18        swap(b[p], b[j]);
19        for (int i = 0; i < n; i++) {
20            if (i != j && (a[i] >> j & U{1})) {
21                a[i] ^= a[j];
22                b[i] ^= b[j];
23            }
24        }
25    }
26    return b;
27 }
28
29 }

```

8.3.3 转置

```

1 namespace gf2 {
2
3 template <typename U>
4 vector<U> transpose(int n, int m, const vector<U>& a) {
5     vector<U> b(m);
6     for (int i = 0; i < n; i++) {
7         for (int j = 0; j < m; j++) {
8             if (a[i] >> j & U{1}) {
9                 b[j] |= U{1} << i;
10            }
11        }
12    }
13    return b;
14 }
15

```

```

16 }

```

8.3.4 GF(2) 线性方程组

```

1 namespace gf2 {
2
3 template <unsigned_integral U>
4 vector<U> solveLinear(int n, int m, vector<U> a, U b) {
5     vector<U> mat(m);
6     U rhs = 0;
7     for (int i = 0; i < n; i++) {
8         U x = a[i], c = b >> i & 1;
9         for (int j = m - 1; j >= 0; j--) {
10             if ((x ^ mat[j]) < x) {
11                 x ^= mat[j];
12                 c ^= rhs >> j & 1;
13             }
14         }
15         if (!x) {
16             if (c) {
17                 return {};
18             }
19             continue;
20         }
21         int k = bit_width(x) - 1;
22         for (int j = k; j < m; j++) {
23             if ((mat[j] ^ x) < mat[j]) {
24                 mat[j] ^= x;
25                 rhs ^= c << j;
26             }
27         }
28         mat[k] = x;
29         rhs |= c << k;
30     }
31     vector<U> res{rhs};
32     mat = transpose(m, m, mat);
33     for (int j = 0; j < m; j++) {
34         if (!(mat[j] >> j & 1)) {
35             res.push_back(mat[j] | (U{1} << j));
36         }
37     }
38     return res;
39 }
40

```

41 }

9 字符串

9.1 Manacher

```

1 vector<int> manacher(string s) {
2     string t = "#";
3     for (auto c : s) {
4         t += c;
5         t += '#';
6     }
7     int n = t.size();
8     vector<int> r(n);
9     for (int i = 0, j = 0; i < n; i++) {
10        if (2 * j - i >= 0 && j + r[j] > i) {
11            r[i] = min(r[2 * j - i], j + r[j] - i);
12        }
13        while (i - r[i] >= 0 && i + r[i] < n && t[i - r[i]] == t[i + r[i]]) {
14            r[i] += 1;
15        }
16        if (i + r[i] > j + r[j]) {
17            j = i;
18        }
19    }
20    return r;
21 }
```

9.2 KMP

```

1 vector<int> kmp(string s) {
2     int n = s.size();
3     vector<int> f(n + 1);
4     for (int i = 1, j = 0; i < n; i++) {
5         while (j && s[i] != s[j]) {
6             j = f[j];
7         }
8         j += (s[i] == s[j]);
9         f[i + 1] = j;
10    }
11    return f;
12 }
13 }
```

9.3 Trie

```

1 struct Trie {
2     struct Node {
3         array<int, 26> nxt;
4         int pass, end;
5
6         Node() {
7             nxt.fill(-1);
8             end = 0;
9             pass = 0;
10        }
11    };
12
13    vector<Node> tr;
14    int tot;
15
16    Trie(int n) {
17        init(n);
18    }
19
20    void init(int n) {
21        tr.resize(n);
22        tot = 0;
23    }
24
25    void insert(string& s) {
26        int p = 0;
27        ++tr[p].pass;
28        for (char& ch : s) {
29            int c = ch - 'a';
30            if (tr[p].nxt[c] == -1) {
31                ++tot;
32                tr[p].nxt[c] = tot;
33            }
34            p = tr[p].nxt[c];
35            ++tr[p].pass;
36        }
37        ++tr[p].end;
38    }
39
40    int count(string& s) {
41        int p = 0;
42        for (char& ch : s) {
43            int c = ch - 'a';
```

```

44         if (tr[p].nxt[c] == -1) {
45             return false;
46         }
47         p = tr[p].nxt[c];
48     }
49     return tr[p].end;
50 }
51 };
52

```

9.4 Z 函数

```

1  vector<int> Z(string s) {
2      int n = s.size();
3      vector<int> z(n + 1);
4      z[0] = n;
5      for (int i = 1, j = 1; i < n; i++) {
6          z[i] = max(0, min(j + z[j] - i, z[i - j]));
7          while (i + z[i] < n && s[z[i]] == s[i + z[i]]) {
8              z[i]++;
9          }
10         if (i + z[i] > j + z[j]) {
11             j = i;
12         }
13     }
14     return z;
15 }
16

```

9.5 AC 自动机

```

1  struct AhoCorasick {
2      static constexpr int ALPHABET = 26;
3      struct Node {
4          int len;
5          int link;
6          array<int, ALPHABET> next;
7          Node() : len{0}, link{0}, next{} {}
8      };
9
10     vector<Node> t;
11
12     AhoCorasick() {
13         init();
14     }
15

```

```

15     void init() {
16         t.assign(2, Node());
17         t[0].next.fill(1);
18         t[0].len = -1;
19     }
20
21     int newNode() {
22         t.emplace_back();
23         return t.size() - 1;
24     }
25
26     int add(const string &a) {
27         int p = 1;
28         for (auto c : a) {
29             int x = c - 'a';
30             if (t[p].next[x] == 0) {
31                 t[p].next[x] = newNode();
32                 t[t[p].next[x]].len = t[p].len + 1;
33             }
34             p = t[p].next[x];
35         }
36         return p;
37     }
38
39     void work() {
40         queue<int> q;
41         q.push(1);
42
43         while (!q.empty()) {
44             int x = q.front();
45             q.pop();
46
47             for (int i = 0; i < ALPHABET; i++) {
48                 if (t[x].next[i] == 0) {
49                     t[x].next[i] = t[t[x].link].next[i];
50                 } else {
51                     t[t[x].next[i]].link = t[t[x].link].next[i];
52                     q.push(t[x].next[i]);
53                 }
54             }
55         }
56     }
57

```

```

58
59     int next(int p, int x) {
60         return t[p].next[x];
61     }
62
63     int link(int p) {
64         return t[p].link;
65     }
66
67     int len(int p) {
68         return t[p].len;
69     }
70
71     int size() {
72         return t.size();
73     }
74 };
75

```

10 杂项

10.1 debug

```

1 using namespace std;
2
3 #define dbg(...) { cerr << "[" << __LINE__ << "]" "; pr(__VA_ARGS__); cerr <<
  '\n'; }
4
5 void pr(bool b) { cerr << (b ? "T" : "F"); }
6 void pr(char c) { cerr << "'" << c << "'"; }
7 void pr(string s) { cerr << "'" << s << "'"; }
8 void pr(const char* s) { cerr << "'" << s << "'"; }
9
10 template <class T>
11 void pr(T&& x) {
12     using U = remove_reference_t<T>;
13     if constexpr (requires { begin(x); }) {
14         bool br = requires { typename U::key_type; };
15         cerr << (br ? '{' : '[');
16         bool f = 0;
17         for (auto&& e : x) {
18             if (exchange(f, 1)) cerr << ",";
19             if constexpr (requires { typename U::mapped_type; }) { pr(e.first);
20                 cerr << ":"; pr(e.second); }
21             else pr(e);
22         }
23     }
24 }
25

```

```

22     cerr << (br ? '}' : ']');
23 } else if constexpr (requires { x.first; x.second; }) {
24     cerr << '('; pr(x.first); cerr << ","; pr(x.second); cerr << ')';
25 } else if constexpr (requires { get<0>(x); }) {
26     cerr << '(';
27     bool f = 0;
28     apply([&](auto... e) { ((cerr << (exchange(f, 1) ? "," : ""),
pr(e)), ...); }, x);
29     cerr << ')';
30 } else if constexpr (requires { x.top(); }) {
31     vector<typename U::value_type> v;
32     while (!x.empty()) v.push_back(x.top()), x.pop();
33     pr(v);
34 } else if constexpr (requires { x.front(); }) {
35     vector<typename U::value_type> v;
36     while (!x.empty()) v.push_back(x.front()), x.pop();
37     pr(v);
38 } else {
39     cerr << x;
40 }
41 }
42
43 template <class A, class... B> requires (sizeof...(B) > 0)
44 void pr(A a, B... b) { pr(a); ((cerr << " ", pr(b)), ...); }
45

```

10.2 template

```

1 using namespace std;
2
3 using i64 = int64_t;
4 using u64 = uint64_t;
5 using u32 = uint32_t;
6
7 using i128 = __int128_t;
8 using u128 = __uint128_t;
9
10 template <typename T>
11 constexpr T inf = numeric_limits<T>::max() / 2;
12 template <>
13 constexpr int inf<int> = 1e9;
14 template <>
15 constexpr i64 inf<i64> = 1e18;
16
17 void solve();
18

```

```

18
19 int main() {
20     ios::sync_with_stdio(false);
21     cin.tie(nullptr);
22
23     int t;
24     cin >> t;
25
26     while (t--) {
27         solve();
28     }
29
30     return 0;
31 }

```

10.3 二分答案

```

1 template <typename T, typename F>
2 T firstTrue(T lo, T hi, F&& pred) {
3     while (lo < hi) {
4         T mid = (lo + hi) / 2;
5         if (pred(mid)) {
6             hi = mid;
7         } else {
8             lo = mid + 1;
9         }
10    }
11    return lo;
12 }
13
14 template <typename T, typename F>
15 T lastTrue(T lo, T hi, F&& pred) {
16     while (hi - lo > 1) {
17         T mid = (lo + hi) / 2;
18         if (pred(mid)) {
19             lo = mid;
20         } else {
21             hi = mid;
22         }
23     }
24     return lo;
25 }

```

10.4 i128

```

1 using i128 = __int128;
2
3 ostream& operator<<(ostream& os, i128 n) {
4     if (n == 0) {
5         return os << 0;
6     }
7     string s;
8     while (n > 0) {
9         s += char('0' + n % 10);
10        n /= 10;
11    }
12    reverse(s.begin(), s.end());
13    return os << s;
14 }
15
16 i128 toi128(const string& s) {
17     i128 n = 0;
18     for (auto c : s) {
19         n = n * 10 + (c - '0');
20     }
21     return n;
22 }
23
24 i128 sqrti128(i128 n) {
25     i128 lo = 0, hi = 1E16;
26     while (lo < hi) {
27         i128 x = (lo + hi + 1) / 2;
28         if (x * x <= n) {
29             lo = x;
30         } else {
31             hi = x - 1;
32         }
33     }
34     return lo;
35 }
36
37 i128 gcd(i128 a, i128 b) {
38     while (b) {
39         a %= b;
40         swap(a, b);
41     }
42     return a;

```

43 }

44

10.5 日期

```
1 const int d[] = {31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};
2
3 bool isLeap(int y) {
4     return y % 400 == 0 || (y % 4 == 0 && y % 100 != 0);
5 }
6
7 int daysInMonth(int y, int m) {
8     return d[m - 1] + (isLeap(y) && m == 2);
9 }
10
11 int getDay(int y, int m, int d) {
12     int ans = 0;
13     for (int i = 1970; i < y; i++) {
14         ans += 365 + isLeap(i);
15     }
16     for (int i = 1; i < m; i++) {
17         ans += daysInMonth(y, i);
18     }
19     ans += d;
20     return (ans + 2) % 7 + 1;
21 }
22
```